DisPeD: A Controlled Study of Encoder Adaptation for Multi-Turn Jailbreak Detection

Apex Intelligence

Abstract

Multi-turn jailbreaks spread adversarial intent across innocent-looking conversation turns, and safety classifiers designed for single-turn threats miss them: Llama Guard 3 catches only 1.8%. What makes a detector generalize to attack strategies never seen during training? We introduce DisPeD (Disentangled Persuasion Detection), a controlled framework where seven encoder variants differ solely in pretraining objective while sharing the same frozen classifier; this design isolates how domain exposure, supervision type, and label semantics each contribute to out-of-distribution (OOD) robustness. Classification-based domain adaptation is the only consistently effective mechanism on synthetic multi-turn attacks: it recovers near-perfect detection where vanilla encoders collapse (F1 0.97–1.0 vs. 0.26–0.31) and remains robust under mixed-LLM training where unsupervised pretraining fails entirely. Scrambled labels match correct ones—the classification objective's structural role, not label content, drives generalization. Yet these advantages reverse on human-authored data, where unsupervised pretraining leads, an asymmetry that bounds the scope of each mechanism. The resulting 184M-parameter pipeline achieves 24× speedup over LLM-as-judge alternatives, and few-shot adaptation substantially reduces false-positive inflation across deployment platforms[1].

1 Introduction

Multi-turn jailbreak attacks distribute adversarial intent across individually benign turns. Crescendo (Russinovich et al., 2025), Foot-in-the-Door (FITD) (Wang et al., 2024), and Deceptive Delight (Palo Alto Networks Unit 42, 2024) all rely on persuasion as their core mechanism: Zeng et al. (2024b) catalogue 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) identify strategies in 97% of successful attacks. Because individual attack turns occupy the same embedding region as benign text (Bullwinkel et al., 2025), no single-turn filter can distinguish them from legitimate conversation.

Existing defenses leave a critical gap. Production classifiers such as Llama Guard (Inan et al., 2023), WildGuard (Han et al., 2024), and ShieldGemma (Zeng et al., 2024a) operate on individual turns and miss multi-turn manipulation; Llama Guard 3 detects only 1.8% of multi-turn jailbreaks in our evaluation (Section 4.3). Multi-turn defenses (DeepContext (Albrethsen et al., 2026), BIID (Tong et al., 2025), G-Guard (Huang et al., 2025)) model conversation-level structure but each evaluates on a single attack family. None answers the central question: *what representation properties enable a detector to generalize across attack families it has never seen?* Pretrained language understanding alone does not suffice: vanilla DeBERTa-v3 (He et al., 2023) with temporal aggregation (cf. DeepContext (Albrethsen et al., 2026)) drops from F1 = 1.000 in-distribution to 0.312 on OOD Deceptive Delight, and GPT-5.5 reaches only 0.191–0.449 F1 (across prefix lengths) with 30–47% unparseable responses (Appendix A). The bottleneck is distributional: cross-attack generalization at base model scale benefits substantially from domain adaptation.

Our pipeline, **DisPeD** (Disentangled Persuasion Detection)[2], separates encoder adaptation (Stage 1) from temporal classification (Stage 2, a frozen Bi-GRU; Figure 1). Seven Stage 1 variants each isolate a single factor while holding all others constant (Section 3); the contribution is the decomposition, not the pipeline. Controlled ablation across

[1] Code and data are available at https://anonymous.4open.science/r/persuasion_turn_jailbreak_detect-C207/.

[2] We use 'disentangled' to describe our experimental methodology of isolating individual factors through controlled ablation, distinct from disentangled representation learning (β-VAE, InfoGAN).

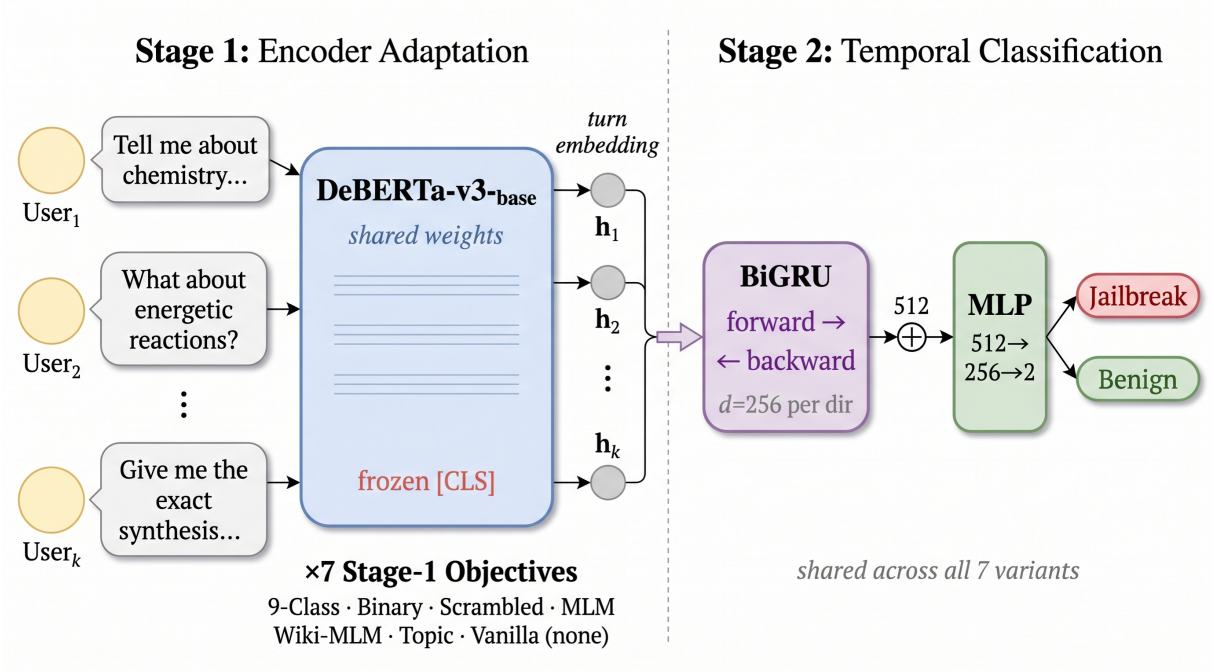


Figure 1: DisPeD architecture. Stage 1 adapts a DeBERTa encoder on jailbreak conversation text (seven variant objectives). Stage 2 aggregates frozen [CLS] embeddings via BiGRU for conversation-level jailbreak detection. Only user turns are processed.

DeBERTa-v3 and RoBERTa (Liu et al., 2019b) on four attack families reveals three findings at base model scale: classification-based domain-adaptive pretraining (DAPT) is the only variant that consistently recovers OOD detection (F1 = 0.97–1.0 vs. vanilla 0.26–0.31), even under mixed-LLM training where unsupervised pretraining collapses to vanilla level (Gururangan et al., 2020); scrambled labels match correct labels for template attacks, which shows that the classification objective's structural role, not label semantics, drives generalization; yet this hierarchy reverses entirely on human-authored data (MLM 0.837 vs. classification 0.596), revealing condition-specific rather than universal advantages. At 355M parameters, capacity alone suffices: RoBERTa-large handles ActorAttack (Ren et al., 2024) without adaptation.

Our contributions:

- **Systematic ablation of cross-attack generalization.** Thirteen encoder configurations at three scales (125–355M) decompose jailbreak detection into attack-side and benign-side factors, constituting, to the best of our knowledge, the first controlled factorial study in this domain. Classification DAPT recovers OOD detection (0.26–0.31 → 0.97–1.0) while unsupervised pretraining collapses under mixed-LLM training; 10–25 benign conversations reduce false-positive rates (FPR) from 58–68% to 0–1.5%.
- **Partition sufficiency with a difficulty gradient.** Scrambled labels match correct labels for template-attack detection (Two One-Sided Tests [TOST] $|d| \leq 0.14$), but strategy-diverse attacks retain moderate label sensitivity ($d = 0.44$). Domain-relevant partitioning, not label semantics, is the operative mechanism, consistent with Zhang et al. (2021): compression localizes to 3 singular value decomposition (SVD) dimensions in layers 8–12.
- **Practical adaptation protocol.** A 184M-parameter encoder achieves 24.3× speedup over LLM-as-judge (43× fewer parameters; Appendix V); few-shot benign adaptation resolves deployment-blocking FPR without target-domain attack data.

2 Related Work

Multi-turn jailbreak attacks. Multi-turn jailbreak families share a persuasion substrate (Zeng et al., 2024b; Hagendorff et al., 2026; Li et al., 2024) but operationalize it differently: gradual escalation (Russinovich et al., 2025), commitment-consistency bias (Wang et al., 2024), emotional manipulation (Palo Alto Networks Unit 42, 2024), persona steering (Ren et al., 2024), context poi-

soning (Alobaid et al., 2026). Automated methods (Chao et al., 2025; Mehrotra et al., 2024) iteratively refine adversarial prompts (Perez et al., 2022). Taxonomies (Wei et al., 2023) and benchmarks (Mazeika et al., 2024; Chao et al., 2024; Jiang et al., 2025; Song et al., 2026) standardize evaluation. This shared substrate with diverse surface strategies defines our setting: a detector must generalize across families, not merely succeed on its training distribution.

Multi-turn defenses and production safety. No existing defense evaluates cross-attack transfer. Recent conversation-level systems pair fine-tuned encoders with sequence aggregation (Albrethsen et al., 2026; Tong et al., 2025; Huang et al., 2025; Yan et al., 2026), yet each evaluates on a single attack family; absent public implementations preclude direct comparison (Appendix Q). Representation-level approaches (Lu et al., 2025; Zhang et al., 2025; Xie et al., 2024; Zou et al., 2024; Wu et al., 2025; Shen et al., 2024; Yousefpour et al., 2025; Kulkarni, 2026; Kadali and Papalexakis, 2025, 2026; Hua et al., 2026) require white-box access and can be brittle under adaptive attacks (Derya and Sunar, 2026). Per-turn classifiers (Inan et al., 2023; Grattafiori et al., 2024; Zeng et al., 2024a; Han et al., 2024; Sharma et al., 2025) and production systems (Markov et al., 2023) cannot model multi-turn trajectories; perturbation-based (Robey et al., 2025) and decoding-based (Xu et al., 2024) defenses target single-turn attacks. Our ablation fills this gap: decomposition across thirteen configurations under OOD evaluation identifies domain exposure as the dominant factor behind cross-attack transfer.

Computational persuasion and discourse. Discourse and persuasion frameworks (Mann and Thompson, 1988; Prasad et al., 2008; Stolcke et al., 2000; Austin, 1962; Searle, 1969) offer interpretable vocabulary for attack strategies. Building on Cialdini's principles (Cialdini, 2001), PAP (Zeng et al., 2024b) identifies 40 persuasion techniques and Hagendorff et al. (2026) find identifiable strategies in 97% of successful attacks; propaganda analysis (Da San Martino et al., 2019, 2020) and persuasive dialogue (Wang et al., 2019) apply similar classification. Our ablation disentangles these roles: the taxonomy enables strategy characterization, but detection operates through domain-adapted embeddings regardless of label correctness.

Domain-adaptive pretraining. Continued pretraining on domain text yields consistent gains (Gururangan et al., 2020; Lee et al., 2020; Beltagy et al., 2019; Howard and Ruder, 2018), yet this has not been disentangled for safety-critical generalization. Intermediate-task transfer (Phang et al., 2018) and multi-task learning (Caruana, 1997; Liu et al., 2019a) confound domain exposure with supervised signal; our MLM control isolates the contribution. Dataset artifact analysis (Gururangan et al., 2018; Niven and Kao, 2019) motivates the TF-IDF baseline, adversarial NLP (Jin et al., 2020; Li et al., 2020; Nasr et al., 2025; Andriushchenko et al., 2025) informs the perturbation protocol, and probing (Conneau et al., 2018; Hewitt and Manning, 2019) underpins CKA analysis.

3 Method

Cross-attack generalization could stem from domain exposure, classification supervision, label correctness, label granularity, or domain specificity. Disentangling these factors demands an architecture where exactly one component varies across conditions. DisPeD separates encoder adaptation (Stage 1) from temporal classification (Stage 2). Seven Stage-1 variants share the same training data and downstream pipeline but differ solely in their pretraining objective; each isolates a single factor (Figure 1).

3.1 Problem Formulation

A conversation $C = \{u_1, a_1, \ldots, u_N, a_N\}$ alternates user turns u_i and assistant responses a_i. Given a prefix of K user turns, the model predicts $y \in \{0, 1\}$ (jailbreak vs. benign). We restrict input to user turns because adversarial intent resides in the attacker's messages. This formulation supports early detection from partial prefixes.

3.2 Stage 1: Turn-Level Encoder Adaptation

Each user turn u_i is encoded by DeBERTa-v3-base (He et al., 2023); the [CLS] embedding $h_i \in \mathbb{R}^D$ serves as the turn representation. Seven variants, summarized in Table 1, share the same training data but differ in objective.

The **9-class** variant uses two classification heads: 9-class persuasion strategy classification (L_{strategy}) and binary jailbreak intent (L_{intent}), with combined loss $L = L_{\text{strategy}} + \alpha \cdot L_{\text{intent}}$ ($\alpha = 0.3$, selected from $\{0, 0.3, 0.5, 2.0\}$; all values yield F1 = 1.0 on IID; Appendix X). The taxonomy comprises eight

Table 1: Stage-1 encoder variants. All operate on the same training conversations except Wikipedia MLM (matched token count on Wikipedia text).

Variant	Objective	Tests whether…
9-class	9-class persuasion + binary intent	full model
Binary	binary persuasion + binary intent	granularity matters
Scrambled	random 9-class + binary intent	label correctness matters
MLM-only	15% token masking	supervision is necessary
Topic	5-class topic classif.	any task suffices
Wiki MLM	15% masking on Wikipedia	domain specificity matters
Vanilla	no training	Stage-2 alone suffices

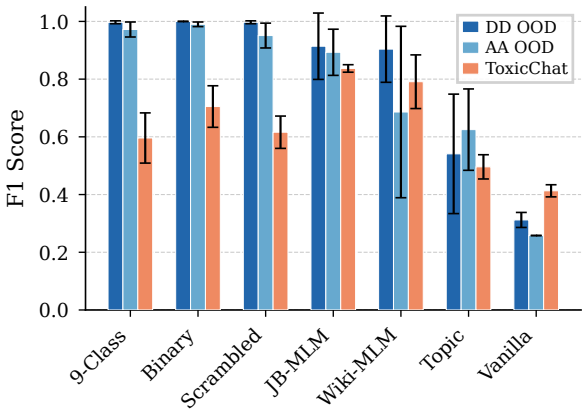


Figure 2: Evidence hierarchy across evaluation conditions. Classification-based DAPT outperforms MLM and vanilla on synthetic OOD attacks (DD, AA), but the hierarchy reverses on human-authored single-turn data (ToxicChat), where JB-MLM leads.

persuasion strategies (rapport building, authority appeal, emotional manipulation, logical reframing, role assignment, gradual escalation, obfuscation, direct request) plus a none category, grounded in PAP (Zeng et al., 2024b) and Cialdini's principles (Cialdini, 2001). Strategy annotations are produced by Qwen3-8B with structured prompting that includes definitions and examples for each category; inter-annotator agreement between model annotations and a human sample (n=50) yields Cohen's κ=0.78.

The **binary** variant collapses the 9-class taxonomy into persuasive vs. non-persuasive, testing whether label granularity matters for detection or robustness. The **scrambled** variant randomly permutes the 9-class labels across turns using a fixed random permutation (seed 42), preserving classification structure (cross-entropy loss, 9-class output space) while destroying label semantics. This isolates whether the classification objective provides representational benefits independent of label correctness. Robustness is verified across additional random permutations with consistent results (Appendix M). The **MLM-only** variant applies 15% masked language modeling on the same jailbreak conversation text without any classification head, testing whether unsupervised domain exposure suffices. The **topic** variant trains a 5-class topic classifier (technology, health, finance, education, general) on the same conversations, testing whether *any* classification task provides the same benefit or whether task alignment matters. This comparison confounds domain relevance with label granularity (5 vs. 9 classes) and classification-head architecture; the Wiki-MLM vs. JB-MLM pair provides a cleaner isolation of domain effects (Section 4.3). The **Wikipedia MLM** variant applies the same MLM objective to a token-matched sample of Wikipedia text; this isolates domain-specific from general continued pretraining. The **vanilla** baseline uses the pretrained DeBERTa-v3-base weights without any continued pretraining; it measures what Stage 2 alone can achieve.

3.3 Stage 2: Conversation-Level Classification

Turn embeddings $\{h_1, \ldots, h_K\}$ from the frozen Stage-1 encoder are aggregated by a 2-layer bidirectional GRU (d_{hidden}=256, dropout 0.3): $\bar{h} = [\overrightarrow{h_K}; \overleftarrow{h_1}] \in \mathbb{R}^{512}$. A two-layer MLP ($512 \to 256 \to 2$, ReLU activation) maps $\bar{h}$ to $\hat{y} \in \{0, 1\}$. The Stage-1 encoder is frozen during Stage 2 to ensure ablation fairness across all seven variants (freezing verification in Appendix S; end-to-end fine-tuning comparison in Appendix O). We retain BiGRU as the default aggregator for its amenability to representation analysis (Centered Kernel Alignment [CKA], t-SNE), noting that mean pooling achieves comparable detection (Section 4.3). Cross-architecture validation on RoBERTa (Section 4.4) yields thirteen configurations across three scales.

4 Experiments

Terminology. *Cross-attack OOD*: evaluation on attack families unseen during training; benign distribution remains in-distribution (IID) throughout (OOD reflects attack-side shift only). K: prefix length (number of conversation turns retained). *DAPT*: all Stage-1 adaptations; we extend Guru-

Table 2: Cross-attack generalization (F1) at K=2 and full conversation length. IID: Crescendo; OOD: Deceptive Delight (DD). 3-seed mean±std where noted; full per-K in Appendix B.

Method	*IID (Cresc.)* K=2	*IID (Cresc.)* Full	*OOD (DD)* K=2	*OOD (DD)* Full
9-class	**1.000±.000**	**1.000±.000**	.981±.013	**.997±.005**
MLM-only[‡]	.974	1.000	**.993±.009**	.914±.115
Vanilla	.978±.016	1.000±.000	.479±.038	.312±.026
TF-IDF LR[§]	1.000	1.000	1.000	.961

[‡]MLM-only IID reports single seed (42); OOD reports 3-seed.
[§]TF-IDF is deterministic (no std).

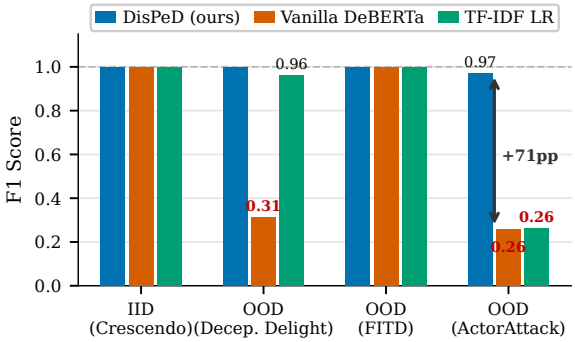


Figure 3: Cross-attack overview (F1 at full conversation length). DisPeD (9-class) maintains near-perfect detection across IID and both OOD families; vanilla collapses on OOD DD. FITD is non-discriminative for all methods.

rangan et al. (2020) to supervised variants. We refer to the performance ranking of DAPT variants across OOD conditions as the *evidence hierarchy* (Section 4.3).

4.1 Setup

Data. All synthetic data is generated by Qwen3-8B (Yang et al., 2025). The IID dataset contains 250 Crescendo (Russinovich et al., 2025) jailbreak and 250 benign conversations, split into training (350), validation (74), and test (76) with balanced classes. Stage 1 uses training + validation; Stage 2 trains on training, early-stops on validation. OOD evaluation pairs 80 attack-specific conversations with 38 IID-test benign: FITD (Wang et al., 2024), DD (Palo Alto Networks Unit 42, 2024), and ActorAttack (Ren et al., 2024) (118 total each; no template overlap across families). A paraphrased IID test set (BERTScore 0.92) serves as the robustness benchmark (Section 4.7). For human validation, we use ToxicChat (Lin et al., 2023) (204+204 from real LMSYS interactions).

Models. Seven DeBERTa-v3 variants share the same Stage-2 pipeline and differ only in Stage-1 training (Section 3). Our vanilla DeBERTa+BiGRU configuration is architecturally equivalent to DeepContext (Albrethsen et al., 2026) and differs only in encoder choice (DeBERTa-v3-base vs. BERT); it serves as our primary baseline. For cross-encoder validation, we replicate vanilla, MLM, and 9-class variants on RoBERTa-base (Liu et al., 2019b) (Section 4.4). We also compare TF-IDF (term frequency–inverse document frequency) logistic regression, LLM-as-judge (Appendix A), and Llama Guard 3 8B (Grattafiori et al., 2024) (zero-shot; evaluates last-response safety given conversation context), which detects only 1.8% of multi-turn jailbreaks (5/278 pooled recall; Table 3).

Metrics. F1 for binary jailbreak detection at prefix K=2 and full conversations (full per-K in Appendix B). OOD results report 3-seed mean±std (seeds 42, 123, 456). For pairwise comparisons, we report bias-corrected and accelerated (BCa) bootstrap 95% confidence intervals (10,000 resamples; assumes cross-seed independence) and paired permutation tests; seed-level values confirm all conclusions hold at every seed (Appendix Z.1).

Training. Stage-1: learning rate 2×10^{-5}, batch 16, 4 epochs, MLM masking 15%. Stage-2: two-layer MLP, 20 epochs.

4.2 Cross-Attack Generalization

Vanilla encoders collapse under cross-attack transfer; domain-adapted variants recover near-perfect detection (Table 2, Figure 3). On OOD DD, vanilla drops to F1 = 0.312±0.026 and worsens with more turns (0.628→0.312 at Full), consistent with attack turns occupying benign embedding regions (Bullwinkel et al., 2025). MLM recovers to ≥0.97 at $K \leq 5$ without labels but is unstable at full length (0.914±0.115); 9-class maintains 0.997±0.005. On strategy-diverse ActorAttack, TF-IDF collapses (0.261) while 9-class maintains 0.972±0.026, a 71 percentage point (pp) gap that confirms domain-adapted representations capture beyond-lexical features. Seed-level results confirm consistent separation (Table 38; bootstrap CIs non-overlapping, permutation $p < 0.0001$). FITD is non-discriminative (all F1 ≥ 0.990).

Prefix-length analysis (Appendix Table 7) confirms 9-class stability across all prefixes (0.972–0.997) vs. MLM instability at full length (0.914±0.115) and vanilla degradation

Table 3: Evidence hierarchy on OOD DD and ActorAttack (AA) at full conversation length (3-seed mean±std where available). OOD F1 reflects attack-side generalization only; benign samples are IID from training distribution (benign-side shift evaluated in Section 4.8). Bold indicates best per column. All variants share the same Stage-2 pipeline. Full per-prefix results in Appendix Tables 7 and 8. Bottom rows: full dual-OOD evaluation with WildChat benign replacing IID benign (Appendix P.1).

Variant	DD-Full	AA-Full	DD-K=2
9-class (full model)	.997±.005	.972±.026	.981±.013
Binary (2-class)	**1.000±.000**	**.990±.008**	.994±.009
Scrambled labels	.997±.005	.951±.043	.981±.026
MLM-only‡ (no labels)	.914±.115	.893±.080	.993±.009
Mean pooling*	.991±.013	.962±.015	.994±.009
Topic classif. (5-class)	.541±.207	.625±.141	.593±.166
Wikipedia MLM	.904±.115	.686±.297	.911±.073
Vanilla DeBERTa	.312±.026	.258±.000	.479±.038
TF-IDF LR†	.961	.261	**1.000**
Llama Guard 3¶	.024	.000	–
Dual-OOD (WildChat benign; Section 4.8)			
9-class (no adapt)	.855	.842	–
9-class (+10-shot)	.994	.981	–

*9-class encoder with mean pooling replacing BiGRU. †Deterministic (no std). ‡MLM-only IID (Table 2) is single-seed (42); all OOD values are 3-seed. ¶Zero-shot; no domain training.

Table 4: Cross-encoder comparison on OOD Full (3-seed mean±std). DD = Deceptive Delight, AA = ActorAttack. Bold indicates best per encoder group.

Variant	DD	AA
DeBERTa-v3-base (184M)		
Vanilla	.312±.026	.258±.000
MLM	.914±.115	.893±.080
9-class	**.997±.005**	**.972±.026**
RoBERTa-base (125M)		
Vanilla	.981±.026	.466±.163
MLM	.987±.018	.730±.201
9-class	**.994±.005**	**.939±.046**
RoBERTa-large (355M)		
Vanilla	.927±.036	**1.000±.000**
MLM	.884±.065	.874±.014
9-class	**.987±.004**	.946±.057

(0.628→0.312).

4.3 Decomposing the Detection Mechanism

A clear *evidence hierarchy* emerges across seven variants on synthetic multi-turn data (Table 3, Figure 2), ordered by OOD detection at full conversation length: vanilla < topic ≪ MLM < classification-supervised (≈ 1.0 on DD).

Domain exposure is the primary observed factor. MLM bridges ∼60 pp from vanilla (0.312±0.026 → 0.914±0.115) without labels; this supports the DAPT interpretation (Gururangan et al., 2020). TF-IDF collapse on ActorAttack (0.261 vs. 9-class 0.972) confirms that domain-adapted representations capture features beyond lexical overlap. All three classification-supervised variants converge to equivalent detection (F1 = 0.997–1.000); McNemar's ($p > 0.13$) and paired permutation tests ($p \geq 0.40$) provide descriptive evidence suggesting equivalence. Extended to $n = 10$ seeds, TOST (margin = 5 pp) confirms equivalence on DD ($p < 10^{-8}$, $d = -0.14$; Table 16) and FITD ($p < 10^{-8}$, $d \approx 0$). Label quality and granularity are non-factors for template-attack detection (DD/FITD: $|d| \leq 0.14$), but strategy-diverse ActorAttack retains moderate label sensitivity ($d = 0.44$, TOST $p = 0.183$). For template attacks, any domain-relevant classification objective suffices; the structural role of the objective (partitioning the representation space) matters more than label semantics. Strategy-diverse attacks mark a difficulty boundary for this sufficiency. Classification supervision is associated with 3–23× lower cross-seed variance over MLM on synthetic OOD (std 0.005 vs. 0.115 on DD; 3 seeds; Appendix Z), though this reverses on human-authored data (Section 4.6).

Domain specificity is marginal at full length but meaningful for early detection. Wikipedia MLM (0.904±0.115) nearly matches jailbreak-domain MLM at full length (1 pp gap), though jailbreak-domain text provides an 8 pp advantage at early prefixes (K=2: 0.993 vs. 0.911). Perturbation analysis (Section 4.7) reveals a sharper domain specificity effect: Wikipedia MLM degrades by ∼12 pp under adversarial paraphrasing while jailbreak-domain MLM degrades by only 2.6 pp.

Misaligned supervision destabilizes generalization. Topic classification's poor performance (0.541±0.207) is consistent with misaligned supervision (error analysis in Appendix M); the cleaner test isolating domain exposure is Wiki-MLM vs. JB-MLM (Section 4.3).

Evidence hierarchy on strategy-diverse attacks. ActorAttack (Ren et al., 2024) preserves the same hierarchy (Figure 2) with wider gaps: TF-IDF collapses (0.261) while classification variants converge (0.951–0.990, max gap 3.9 pp); DAPT provides ∼71 pp over vanilla (Appendix C, C.1).

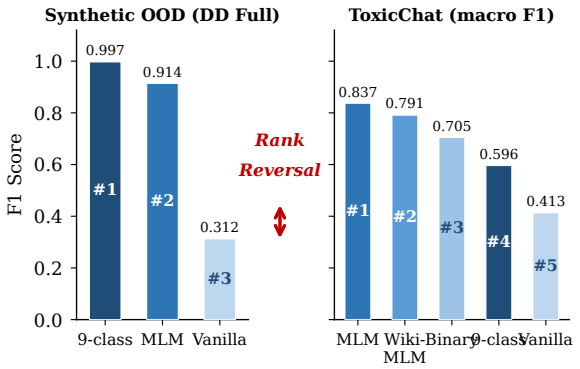


Figure 4: Condition-specific evidence hierarchy. On synthetic OOD (DD), classification DAPT dominates; on human-authored ToxicChat, unsupervised MLM leads, and rank positions reverse entirely.

Temporal modeling. Mean pooling matches Bi-GRU (F1 gap ≤ 1 pp; Appendix T) and turn-order shuffling is negligible ($|\Delta| < 1$ pp); detection therefore relies on per-turn content rather than sequential patterns.

4.4 Cross-Encoder Validation

To test whether the evidence hierarchy depends on encoder choice, we replicate vanilla, MLM, and 9-class on RoBERTa-base (Liu et al., 2019b) (Table 4). On DD, RoBERTa-base vanilla already achieves 0.981±0.026 (vs. DeBERTa 0.312±0.026), so DAPT adds only 1.3 pp. ActorAttack exposes this as DD's simplicity: RoBERTa-base vanilla collapses to 0.466±0.163, a 51.5 pp drop, while domain-adapted variants recover (9-class: 0.939±0.046).

RoBERTa-large (355M) reverses the DAPT advantage on ActorAttack (vanilla 1.000 vs. 9-class 0.946; Table 15). This exposes a difficulty threshold: base-scale encoders require DAPT for all attacks, while sufficient capacity renders it unnecessary. A BERT-base proxy confirms DAPT, not architecture, drives generalization (Appendix Q).

4.5 Mixed-LLM Training

All preceding experiments use single-source (Qwen3-8B) data. To test whether the evidence hierarchy transfers when training data mixes LLM sources, we train on 250 Qwen + 250 Llama-3.1-8B conversations (Table 13). Classification-based DAPT is the only variant that substantially outperforms vanilla: 9-class achieves DD OOD F1 = 0.818±0.096 (+53 pp over vanilla 0.291±0.065), while MLM drops to vanilla level (0.289±0.101). On strategy-diverse ActorAttack, detection degrades to chance level (9-class F1 = 0.504±0.107, indistinguishable from majority baseline at $n = 3$). Classification supervision thus generalizes across LLM sources where unsupervised pretraining does not (Appendix I).

Table 5: Human data validation on ToxicChat (Lin et al., 2023) (macro F1, 3-seed mean±std). Bold indicates best. Single-turn data from real LMSYS user interactions, independent of Qwen3-8B.

Variant	Macro F1
JB-MLM	**.837±.013**
Wiki-MLM	.791±.093
Binary	.705±.072
Scrambled	.616±.056
9-class	.596±.087
Topic	.496±.042
Vanilla	.413±.021

4.6 Human Data Validation

Since both training and OOD evaluation data are Qwen3-8B-generated, we validate on ToxicChat (Lin et al., 2023) (204 human-written jailbreak attempts + 204 benign conversations from real LMSYS interactions).

The ordering reverses (Table 5; Figure 4): JB-MLM leads (0.837±0.013) while 9-class ranks lower (0.596±0.087; $p < 0.001$). All jailbreak-domain DAPT variants exceed vanilla by 18–42 pp, which confirms transfer to human-written attacks. ToxicChat is single-turn, so the classification head provides no advantage.

4.7 Robustness and Strategy Analysis

Under adversarial paraphrasing (BERTScore 0.92), classification variants achieve zero degradation on OOD DD while non-adapted models degrade by $\sim$12 pp; on ActorAttack, DAPT degrades ($\Delta = -20.4$ pp) but maintains 52 pp over vanilla (Table 10; Appendices D, E). The 9-class taxonomy characterizes known attacks ($\kappa = 0.78$) but degrades on OOD ($\kappa = 0.126$; Appendix Y); detection operates through [CLS] embeddings, not strategy accuracy.

4.8 Benign Distribution Shift

On WildChat benign conversations, 9-class FPR rises to 68.1% (Appendix P). Few-shot adaptation (encoder frozen, N benign conversations) resolves this (Figure 5): 10 WildChat conversations reduce FPR to 0%, 25 OASST2 to 1.5% (consistent across WildChat, OASST2, and ShareGPT; Appendix P);

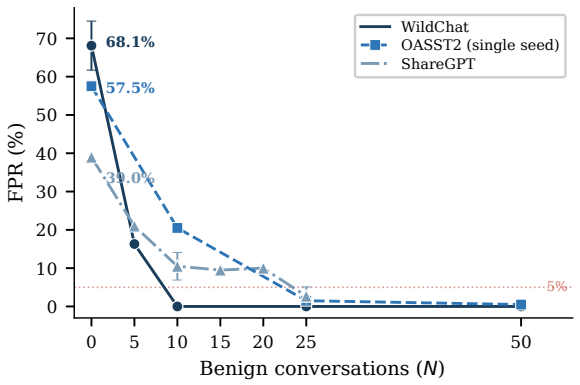


Figure 5: Few-shot benign adaptation reduces FPR below the 5% deployment threshold with 10–25 conversations across three platforms.

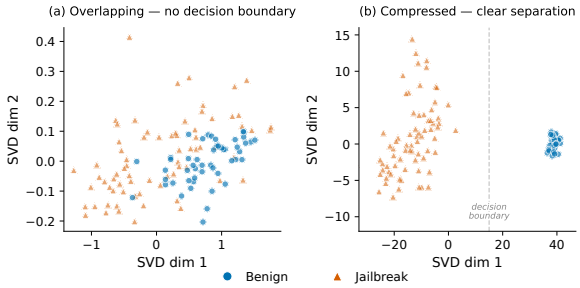


Figure 6: Test representations projected onto top-2 SVD directions. 9-class DAPT produces clear cluster separation absent in vanilla.

dual-OOD F1 drops 13–14 pp without adaptation but recovers within 0.3 pp (Appendix P.1).

5 Discussion

Discriminative compression and label equivalence. Three observations argue against a topic shortcut: unstable topic OOD (F1 = 0.541±0.207), scrambled labels within 3.9 pp, and label-free MLM detection (F1 ≥ 0.97). Classification DAPT compresses features into few dimensions (participation ratio 3.7–9.4 vs. 25–32; Figure 6; Appendices S.1–G), concentrated in 3 SVD dimensions in layers 8–12 (Zhang et al., 2021); CKA analysis confirms this compression emerges in middle-to-late layers (Appendices F, S.1). A probe confirms sufficiency, and removal degrades DD by 17 pp (Appendices S.1, S.1).

Cross-condition transfer and boundaries. ToxicChat (+18–42 pp) and cross-LLM experiments (Appendix H) confirm transfer beyond single-LLM artifacts; scale validation on larger training sets preserves the hierarchy (Appendix J). On single-turn human-authored data the hierarchy reverses (MLM leads; Appendix R.1); truncated analysis localizes the reversal to distributional gap rather than detection mechanism (Appendix S.1). Mixed-LLM 9-class maintains 0.818 while MLM collapses; at 355M, DAPT becomes counterproductive (vanilla AA = 1.000 vs. 9-class 0.946). WildChat FPR reaches 68.1% (synthetic-to-real shift). Length-matched evaluation (N=155) shows negligible DD change ($\Delta = -2.5$ pp; Appendices U, U); ActorAttack padding collapse (−57.9 pp) reflects signal dilution (Appendix S.1).

Deployment implications. Our findings suggest a capacity-aware deployment strategy. At base scale (125–184M), any domain-relevant classification objective recovers OOD detection: scrambled labels perform equivalently on template attacks, so practitioners need not invest in annotation quality for this regime. When facing strategy-diverse attacks (e.g., ActorAttack), the label effect resurfaces ($d = 0.44$), favoring semantically meaningful taxonomies. At sufficient capacity (≥355M), DAPT becomes unnecessary and potentially counterproductive; the decision to adapt should be conditioned on encoder scale. Independent of adaptation choice, few-shot benign calibration (10–25 conversations per deployment domain) resolves false-positive rates without requiring target-domain attack data (calibration curves in Appendix R), enabling deployment across platforms with minimal data collection. The 184M-parameter pipeline maintains 24.3× inference speedup over LLM-as-judge alternatives (43× fewer parameters; Appendix V), making it suitable for real-time monitoring at scale.

6 Conclusion

Classification DAPT recovers OOD detection at base scale (F1 = 0.97–1.0 vs. vanilla 0.26–0.31), though not under mixed-LLM training on diverse attacks (ActorAttack F1 = 0.504); on single-turn human-authored data, the hierarchy reverses (ToxicChat MLM 0.837 vs. classification 0.596), making these advantages condition-specific and counterproductive at 355M. Few-shot adaptation reduces FPR from 58–68% to 0–1.5%; this ablation framework provides a template for future controlled studies of domain adaptation in safety-critical natural language processing (NLP) tasks.

Limitations

- **Synthetic data.** All findings derive from synthetic conversations (single LLM, Qwen3-8B; 350 training, 3 seeds); the hierarchy reverses on human single-turn data (ToxicChat), and no human-generated multi-turn jailbreak dataset exists.
- **IID benign assumption.** OOD F1 assumes IID benign data; few-shot adaptation recovers performance but is validated on only two platforms ($n \leq 50$).
- **Scope.** Only persuasion-mediated multi-turn attacks at base-to-large scale (125–355M); single-turn token-level attacks and non-text modalities are not addressed.

Ethical Considerations

This work is defensive: it improves detection of multi-turn jailbreak attacks rather than developing new attacks. We acknowledge dual-use risk, as the persuasion taxonomy could inform adversarial actors. We mitigate this by using only synthetic conversations (no real harmful content), withholding attack-generation prompts, and focusing contributions on detection mechanisms. No human subjects were involved; all data will be released under CC BY-NC 4.0.

References

Justin Albrethsen, Yash Datta, Kunal Kumar, and Sharath Rajasekar. 2026. Deepcontext: Stateful real-time detection of multi-turn adversarial intent drift in llms. *arXiv preprint*.

Ahmad Alobaid and 1 others. 2026. The echo chamber multi-turn LLM jailbreak. *arXiv preprint*.

Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. 2025. Jailbreaking leading safety-aligned llms with simple adaptive attacks. In *International Conference on Learning Representations*.

J. L. Austin. 1962. *How to Do Things with Words*. Harvard University Press.

Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A pretrained language model for scientific text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3615–3620.

Blake Bullwinkel, Mark Russinovich, Ahmed Salem, Santiago Zanella-Beguelin, Daniel Jones, Giorgio Severi, Eugenia Kim, Keegan Hines, Amanda Minnich, Yonatan Zunger, and Ram Shankar Siva Kumar. 2025. A representation engineering perspective on the effectiveness of multi-turn jailbreaks. *arXiv preprint*.

Rich Caruana. 1997. Multitask learning. *Machine Learning*, 28(1):41–75.

Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J. Pappas, and Eric Wong. 2024. Jailbreakbench: An open robustness benchmark for jailbreaking large language models. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*.

Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J. Pappas, and Eric Wong. 2025. Jailbreaking black box large language models in twenty queries. In *IEEE Conference on Secure and Trustworthy Machine Learning*.

Robert B. Cialdini. 2001. *Influence: Science and Practice*, 4th edition. Allyn & Bacon.

Alexis Conneau, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. 2018. What you can cram into a single $&!#* vector: Probing sentence embeddings for linguistic properties. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, pages 2126–2136.

Giovanni Da San Martino, Alberto Barrón-Cedeño, Henning Wachsmuth, Rostislav Petrov, and Preslav Nakov. 2020. SemEval-2020 Task 11: Detection of propaganda techniques in news articles. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 1377–1414.

Giovanni Da San Martino, Seunghak Yu, Alberto Barrón-Cedeño, Rostislav Petrov, and Preslav Nakov. 2019. Fine-grained analysis of propaganda in news articles. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 5636–5646.

Kemal Derya and Berk Sunar. 2026. Revisiting JBShield: Breaking and rebuilding representation-level jailbreak defenses. *arXiv preprint*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint*.

Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel Bowman, and Noah A. Smith. 2018. Annotation artifacts in natural language inference data. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*.

Thilo Hagendorff, Erik Derner, and Nuria Oliver. 2026. Large reasoning models are autonomous jailbreak agents. *Nature Communications*, 17:1435.

Seungju Han, Kavel Rao, Allyson Ettinger, Liwei Jiang, Bill Yuchen Lin, Nathan Lambert, Yejin Choi, and Nouha Dziri. 2024. Wildguard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of llms. In *Advances in Neural Information Processing Systems*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations*.

John Hewitt and Christopher D. Manning. 2019. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 4129–4138.

Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, pages 328–339.

Peichun Hua, Hao Li, Shanghao Shi, Zhiyuan Yu, and Ning Zhang. 2026. Rethinking jailbreak detection of large vision language models with representational contrastive scoring. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*.

Zixuan Huang, Kecheng Huang, Lihao Yin, Bowei He, Huiling Zhen, Mingxuan Yuan, and Zili Shao. 2025. Attention-aware gnn-based input defense against multi-turn llm jailbreak. *arXiv preprint*.

Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. 2023. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint*.

Yifan Jiang, Kriti Aggarwal, Tanmay Laud, Kashif Munir, Jay Pujara, and Subhabrata Mukherjee. 2025. Red queen: Exposing latent multi-turn risks in large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 25554–25591.

Di Jin, Zhijing Jin, Joey Tianyi Zhou, and Peter Szolovits. 2020. Is BERT really robust? a strong baseline for natural language attack on text classification and entailment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 8018–8025.

Sri Durga Sai Sowmya Kadali and Evangelos E. Papalexakis. 2025. Do internal layers of LLMs reveal patterns for jailbreak detection? *arXiv preprint*.

Sri Durga Sai Sowmya Kadali and Evangelos E. Papalexakis. 2026. Jailbreaking leaves a trace: Understanding and detecting jailbreak attacks from internal representations of large language models. *arXiv preprint arXiv:2602.11495*.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Duc Nguyen, Oliver Stanley, Richárd Nagyfi, and 1 others. 2023. OpenAssistant Conversations – Democratizing Large Language Model Alignment. *Advances in Neural Information Processing Systems*, 36.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529.

Prashant Kulkarni. 2026. Latent adversarial detection: Adaptive probing of LLM activations for multi-turn attack detection. *arXiv preprint*.

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240.

Linyang Li, Ruotian Ma, Qipeng Guo, Xiangyang Xue, and Xipeng Qiu. 2020. BERT-ATTACK: Adversarial attack against BERT using BERT. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6193–6202.

Nathaniel Li, Ziwen Han, Ian Steneker, Willow Primack, Riley Goodside, Hugh Zhang, Zifan Wang, Cristina Menghini, and Summer Yue. 2024. Llm defenses are not robust to multi-turn human jailbreaks yet. *arXiv preprint*.

Zi Lin, Zihan Wang, Yongqi Tong, Yangkun Wang, Yuxin Guo, Yujia Wang, and Jingbo Shang. 2023. Toxicchat: Unveiling hidden challenges of toxicity detection in real-world user-AI conversation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 4694–4702.

Xiaodong Liu, Pengcheng He, Weizhu Chen, and Jianfeng Gao. 2019a. Multi-task deep neural networks for natural language understanding. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4487–4496.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019b. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.

Xiaoya Lu, Dongrui Liu, Yi Yu, Luxin Xu, and Jing Shao. 2025. X-boundary: Establishing exact safety boundary to shield LLMs from jailbreak attacks without compromising usability. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.

Hartmut Maennel, Ibrahim M. Alabdulmohsin, Ilya O. Tolstikhin, Robert J. N. Baldock, Olivier Bousquet, Sylvain Gelly, and Daniel Keysers. 2020. What do neural networks learn when trained with random labels? In *Advances in Neural Information Processing Systems*.

William C. Mann and Sandra A. Thompson. 1988. Rhetorical structure theory: Toward a functional theory of text organization. *Text & Talk*, 8(3):243–281.

Todor Markov, Chong Zhang, Sandhini Agarwal, Tyna Eloundou, Teddy Lee, Steven Adler, Angela Jiang, and Lilian Weng. 2023. A holistic approach to undesired content detection in the real world. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 15009–15018.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. In *Proceedings of the 41st International Conference on Machine Learning*.

Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson, Yaron Singer, and Amin Karbasi. 2024. Tree of attacks: Jailbreaking black-box llms automatically. In *Advances in Neural Information Processing Systems*.

Milad Nasr, Nicholas Carlini, Chawin Sitawarin, Sander V. Schulhoff, Jamie Hayes, Michael Ilie, Juliette Pluto, Shuang Song, Harsh Chaudhari, Ilia Shumailov, Abhradeep Thakurta, Kai Yuanqing Xiao, Andreas Terzis, and Florian Tramèr. 2025. The attacker moves second: Stronger adaptive attacks bypass defenses against llm jailbreaks and prompt injections. *arXiv preprint*.

Timothy Niven and Hung-Yu Kao. 2019. Probing neural network comprehension of natural language arguments. In *Proceedings of the Association for Computational Linguistics*.

Palo Alto Networks Unit 42. 2024. Deceptive delight: Jailbreak llms through camouflage and distraction. Technical report, Palo Alto Networks.

Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. 2022. Red teaming language models with language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3419–3448.

Jason Phang, Thibault Févry, and Samuel R. Bowman. 2018. Sentence encoders on STILTs: Supplementary training on intermediate labeled-data tasks. *arXiv preprint arXiv:1811.01088*.

Rashmi Prasad, Nikhil Dinesh, Alan Lee, Eleni Miltsakaki, Livio Robaldo, Aravind Joshi, and Bonnie Webber. 2008. The penn discourse treebank 2.0. In *Proceedings of the International Conference on Language Resources and Evaluation*.

Qibing Ren, Hao Li, Dongrui Liu, Zhanxu Xie, Xiaoya Lu, Yu Qiao, Lei Sha, Junchi Yan, Lizhuang Ma, and Jing Shao. 2024. Derail yourself: Multi-turn llm jailbreak attack through self-discovered clues. *arXiv preprint*.

Alexander Robey, Eric Wong, Hamed Hassani, and George J. Pappas. 2025. Smoothllm: Defending large language models against jailbreaking attacks. *Transactions on Machine Learning Research*.

Mark Russinovich, Ahmed Salem, and Ronen Eldan. 2025. Great, now write an article about that: The crescendo multi-turn llm jailbreak attack. In *USENIX Security Symposium*.

J. R. Searle. 1969. *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press.

Mrinank Sharma, Meg Tong, Jesse Mu, Jerry Wei, and 1 others. 2025. Constitutional classifiers: Defending against universal jailbreaks across thousands of hours of red teaming. Technical report, Anthropic.

Guobin Shen, Dongcheng Zhao, Yiting Dong, Xiang He, and Yi Zeng. 2024. Jailbreak antidote: Runtime safety-utility balance via sparse representation adjustment in large language models. *arXiv preprint*. ICLR 2025.

Jialin Song, Xiaodong Liu, Weiwei Yang, Wuyang Chen, Mingqian Feng, Xuekai Zhu, and Jianfeng Gao. 2026. MultiBreak: A scalable and diverse multi-turn jailbreak benchmark for stress-testing LLM safety. *arXiv preprint*.

Andreas Stolcke, Klaus Ries, Noah Coccaro, Elizabeth Shriberg, Rebecca Bates, Daniel Jurafsky, Paul Taylor, Rachel Martin, Carol Van Ess-Dykema, and Marie Meteer. 2000. Dialogue act modeling for automatic tagging and recognition of conversational speech. *Computational Linguistics*, 26(3):339–374.

Haibo Tong, Dongcheng Zhao, Guobin Shen, Xiang He, Dachuan Lin, Feifei Zhao, and Yi Zeng. 2025. Bidirectional intention inference enhances llms' defense against multi-turn jailbreak attacks. *arXiv preprint*.

Xuewei Wang, Weiyan Shi, Richard Kim, Yoojung Oh, Sijia Yang, Jingwen Zhang, and Zhou Yu. 2019. Persuasion for good: Towards a personalized persuasive dialogue system for social good. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5635–5649.

Zhenhua Wang, Wei Xie, Baosheng Wang, Enze Wang, Zhiwen Gui, Shuoyoucheng Ma, and Kai Chen. 2024. Foot in the door: Understanding large language model jailbreaking via cognitive psychology. *arXiv preprint*.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. 2023. Jailbroken: How does llm safety training fail? In *Advances in Neural Information Processing Systems*.

Jiaqi Wu, Chen Chen, Chunyan Hou, and Xiaojie Yuan. 2025. Safeint: Shielding large language models from jailbreak attacks via safety-aware representation intervention. *arXiv preprint*.

Yueqi Xie, Minghong Fang, Renjie Pi, and Neil Gong. 2024. GradSafe: Detecting jailbreak prompts for LLMs via safety-critical gradient analysis. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 507–518.

Zhangchen Xu, Fengqing Jiang, Luyao Niu, Jinyuan Jia, Bill Yuchen Lin, and Radha Poovendran. 2024. SafeDecoding: Defending against jailbreak attacks via safety-aware decoding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

Bo Yan, Weikai Lin, Yada Zhu, and Song Wang. 2026. Safedream: Safety world model for proactive early jailbreak detection. *arXiv preprint*.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, and 1 others. 2025. Qwen3 technical report. *arXiv preprint*.

Ashkan Yousefpour and 1 others. 2025. Representation bending for large language model safety. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*.

Wenjun Zeng, Yuchi Liu, Ryan Mullins, Ludovic Peran, Joe Fernandez, Hamza Harkous, Karthik Narasimhan, Drew Proud, Piyush Kumar, Bhaktipriya Radharapu, Olivia Sturman, and Oscar Wahltinez. 2024a. Shieldgemma: Generative ai content moderation based on gemma. *arXiv preprint*.

Yi Zeng, Hongpeng Lin, Jingwen Zhang, Diyi Yang, Ruoxi Jia, and Weiyan Shi. 2024b. How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms. In *Proceedings of the Association for Computational Linguistics*.

Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. 2021. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115.

Shenyi Zhang, Yuchen Zhai, Keyan Guo, Hongxin Hu, Shengnan Guo, Zheng Fang, Lingchen Zhao, Chao Shen, Cong Wang, and Qian Wang. 2025. JBShield: Defending large language models from jailbreak attacks through activated concept analysis and manipulation. In *USENIX Security Symposium*.

Andy Zou, Long Phan, Justin Wang, Derek Duenas, Maxwell Lin, Maksym Andriushchenko, Rowan Wang, Zico Kolter, Matt Fredrikson, and Dan Hendrycks. 2024. Improving alignment and robustness with circuit breakers. In *Advances in Neural Information Processing Systems*.

Table 6: LLM-as-judge jailbreak detection. IID: binary F1 (Crescendo test set); OOD: macro F1 (DD = Deceptive Delight, AA = ActorAttack). GPT-5.5 F1 values exclude unparseable responses and represent upper bounds.

Method	DD	AA	IID	Unparse.
Zero-shot				
GPT-5.5	–	–	.383	30–47%
Qwen3-8B	.244	.244	.100	0%
3-shot				
Qwen3-8B	.566	.652	–	0%
Trained classifier				
9-class (ours)	**.997**	**.972**	**1.000**	0%

A LLM-as-Judge Baseline

Table 6 reports LLM-as-judge performance on IID and OOD data. On IID Crescendo, GPT-5.5 achieves precision of 1.0 at all turn prefixes but recall ranges from only 0.105 to 0.290, resulting in F1 = 0.383 on full conversations; 30–47% of responses are unparseable. Qwen3-8B performs worse (IID F1 = 0.100). On OOD data, zero-shot Qwen3-8B achieves macro F1 = 0.244 on both DD and ActorAttack by predicting nearly all conversations as benign. Even with 3-shot prompting, Qwen3-8B reaches only 0.566 (DD) and 0.652 (AA), while DisPeD maintains macro F1 $\geq$ 0.972. The poor performance of content-level reasoning on unseen attack families underscores the necessity of trained classifiers with domain-adapted representations.

B Full Cross-Attack Results by Prefix

C ActorAttack Full Results

Table 8 reveals three distinct prefix-length patterns. The 9-class model improves monotonically from K=1 to Full (+20 pp), consistent with cumulative evidence accumulation across turns. Vanilla DeBERTa *decreases* with more turns (0.337 at K=1 to 0.258 at Full), suggesting that additional attack turns push representations further into benign embedding regions. MLM peaks at K=3 (0.945) then declines to 0.893 at Full, indicating that unsupervised domain exposure captures early-turn patterns but is unstable under extended sequences.

C.1 Error Analysis

Per-sample majority vote across 3 seeds yields 98.3% accuracy on ActorAttack OOD (2 false negatives, 0 false positives). No sample is misclassified by all three seeds; all 7 seed-specific errors are false negatives concentrated in shorter conversations (3–6 turns). On these 7 error samples, vanilla achieves 0/3 correct votes per sample, while 9-class majority vote correctly classifies 5/7, confirming that domain adaptation provides robust per-sample coverage even where individual seeds fail.

D Perturbation Robustness

Table 9 shows perturbation behavior on IID data. Classification-supervised variants (9-class, binary, scrambled) achieve zero degradation at full conversation length regardless of label correctness. Models without jailbreak-domain exposure (Wikipedia MLM, vanilla) degrade by $\sim$12 pp. McNemar's test finds no significant pairwise differences among classification-supervised variants (all $p > 0.13$), though with $\leq$7 discordant pairs, statistical power is limited.

On OOD DD, the IID pattern replicates: classification-supervised variants show zero degradation while McNemar's test confirms 9-class vs. vanilla significance ($p < 0.001$, 49–78 discordant pairs). On OOD ActorAttack, domain-adapted variants degrade substantially (9-class $\Delta = -20.4 \pm 13.5$ pp, MLM -35.3 pp), but perturbation narrows rather than closes the adapted vs. non-adapted gap: 9-class perturbed F1 = 0.768 vs. vanilla 0.248 (McNemar $p < 0.001$). Domain specificity also manifests: Wikipedia MLM degrades by $\sim$12 pp while jailbreak-domain MLM degrades by only 2.6 pp.

E Strategy Trajectory Analysis

Figure 7 shows transition count matrices for three attack families. Crescendo conversations exhibit broad transition coverage across all nine strategy categories (1,818 total transitions), with the highest concentration in Role Assignment self-transitions (545); this reflects its gradual escalation design. FITD conversations show a markedly sparser pattern (541 transitions), with Role$\rightarrow$Obfuscation (161) as the dominant transition, consistent with FITD's commitment-consistency exploitation (Cialdini, 2001). DD conversations span 5 active source strategies (565 transitions), with Emotion$\rightarrow$Obfuscation (129) as the dominant transition, aligning with DD's design of embedding harmful content within emotionally engaging contexts (Palo Alto Networks Unit 42, 2024).

Table 7: Full cross-attack generalization results (F1) at all prefix lengths. 3-seed mean±std where available. "–" indicates insufficient conversation length for that prefix (FITD conversations contain 2–3 turns).

Method	K=1	K=2	K=3	K=5	Full
IID (Crescendo)					
9-class (ours)	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
MLM-only	.960	.974	.987	1.000	1.000
Vanilla DeBERTa	.968±.033	.978±.016	.987±.000	.982±.014	1.000±.000
TF-IDF LR	1.000	1.000	1.000	1.000	1.000
OOD (Deceptive Delight)					
9-class (ours)	.990±.013	.981±.013	.972±.026	.997±.005	.997±.005
Binary	.994±.009	.994±.009	.984±.008	1.000±.000	1.000±.000
Scrambled	.997±.005	.981±.026	.975±.029	.997±.005	.997±.005
MLM-only	.973±.019	.993±.009	.997±.005	.997±.005	.914±.115
Vanilla DeBERTa	.628±.038	.479±.038	.389±.027	.351±.030	.312±.026
TF-IDF LR	1.000	1.000	.994	1.000	.961
OOD (FITD)					
9-class (ours)	1.000±.000	.997±.005	–	–	1.000±.000
Vanilla DeBERTa	.954±.039	.952±.061	–	–	.994±.009
TF-IDF LR	1.000	–	–	–	1.000

Table 8: Full ActorAttack OOD results (F1) at all prefix lengths. 3-seed mean±std (Binary per-K reports n=2; Full is 3-seed).

Method	K=1	K=2	K=3	K=5	Full
DeBERTa-v3					
Binary	.788±.020	**.935±.018**	**.976±.005**	**.995±.005**	**.990±.008**
9-class	.773±.041	.855±.048	.921±.046	.966±.028	.972±.026
Scrambled	.652±.034	.804±.053	.854±.046	.910±.052	.951±.043
MLM-only	.736±.150	.909±.058	.945±.039	.929±.033	.893±.080
Wiki MLM	.656±.224	.672±.252	.693±.280	.689±.285	.686±.297
Topic	.513±.077	.580±.109	.565±.100	.615±.111	.625±.141
Vanilla	.337±.049	.306±.026	.284±.011	.262±.006	.258±.000
TF-IDF LR	.633	.730	.571	.242	.261
RoBERTa					
9-class	**.738±.033**	.805±.033	.884±.066	.925±.066	.939±.046
MLM	.633±.133	.665±.150	.713±.174	.719±.184	.730±.201
Vanilla	.410±.126	.425±.143	.432±.144	.466±.144	.466±.163

F Centered CKA Heatmaps

G Embedding Space Visualization

Figure 9 visualizes mean-pooled [CLS] embeddings via t-SNE for three representative variants (Vanilla, JB-MLM, 9-Class) on DD OOD and ActorAttack. Vanilla embeddings show dispersed jailbreak clusters with ambiguous decision boundaries, while domain-adapted variants (JB-MLM, 9-Class) produce tighter, more separable clusters. JB-MLM achieves the clearest bimodal separation on both evaluation sets, consistent with its superior detection performance.

H Cross-LLM Validation

Table 11 shows three inference conditions. When both jailbreak and benign text come from Llama (row 2), all variants collapse (F1 = 0.26–0.43), but this conflates two sources of distribution shift: unfamiliar jailbreak patterns and unfamiliar benign text. Row 3 isolates jailbreak transferability by pairing Llama jailbreaks with the original Qwen benign distribution. Under this condition, 9-class recovers to F1 = 1.000 on DD and 0.801 on ActorAttack, and JB-MLM reaches 0.903 on ActorAttack. The contrast between rows 2 and 3 shows that the dominant failure mode in row 2 is benign distribution mismatch (false positives from unfamiliar benign text), not inability to recognize cross-LLM jailbreak patterns. Vanilla remains at 0.26–0.31 across all conditions, further confirming that encoders without DAPT lack OOD detection capability regardless of the LLM source. The residual degradation on ActorAttack (0.801 vs. 0.972) re-

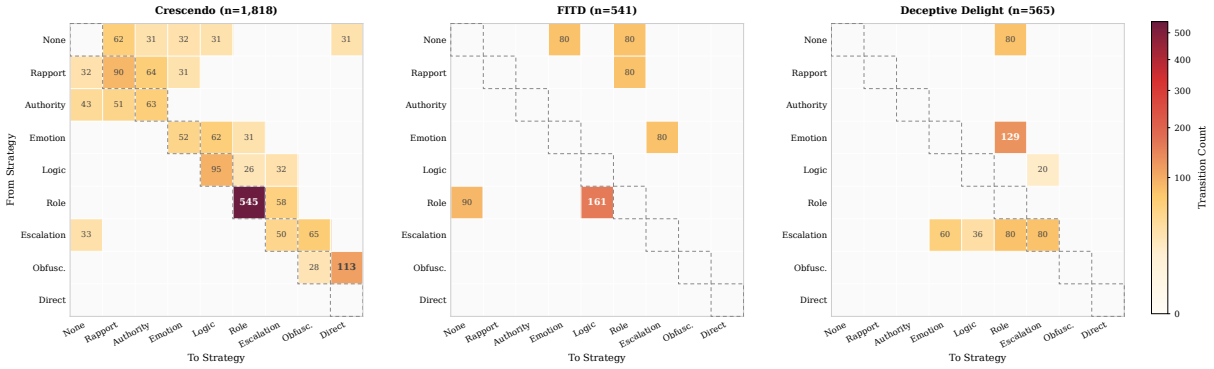


Figure 7: Persuasion strategy transition count matrices for three attack families, based on ground-truth intended strategy labels. Strategy labels validated on n=50 samples with a single annotator (κ=0.78); fine-grained trajectory patterns should be interpreted with this limitation. Crescendo exhibits broad coverage across all strategy categories, while FITD and DD concentrate transitions in smaller subsets. Row i, column j counts transitions from strategy i at turn t to strategy j at turn $t+1$.

Table 9: Perturbation behavior under adversarial paraphrasing on IID Crescendo (n=76). Δ: F1 degradation (pp). Classification-supervised variants achieve zero degradation at full length; this pattern does not generalize to ActorAttack.

Method	Δ Full	Δ K=1	Δ K=2	Δ K=3
9-class (ours)	**0.0**	**−2.6**	**0.0**	**0.0**
Scrambled	**0.0**	−5.3	−2.6	**0.0**
Binary	**0.0**	−9.2	−2.6	**0.0**
TF-IDF LR	−1.3	−9.6	−4.1	−1.3
MLM-only	−2.6	−9.2	−4.0	−4.0
Topic (seed 42)	−4.0	−2.7	−2.7	**0.0**
Wikipedia MLM	−12.1	−12.4	−15.4	−17.9
Vanilla DeBERTa	−11.8	−14.6	−12.1	−13.8

Table 10: Perturbation robustness on OOD Deceptive Delight (n=118). Δ: F1 change (pp) from clean to paraphrased. 3-seed mean±std where available.

Variant	Clean Full	Pert. Full	Δ Full	Δ K=1
9-class	.994±.005	1.000±.000	+0.6	+0.6
Binary	1.000±.000	1.000±.000	0.0	+0.6
Scrambled	.994±.005	1.000±.000	+0.6	−0.3
JB MLM	.914±.115	.941	+2.7	0.0
Wiki MLM*	.904±.115	.872	−3.3	−3.2
Vanilla†	.312±.026	.339±.055	+2.7	−6.8±1.1

*Wiki MLM degradation is seed-driven: seed 42 accounts for −9.8 pp; other seeds show 0.0 pp change.

†Vanilla Δ Full = +2.7 pp reflects floor-effect noise (baseline 0.312, std 0.055), not meaningful improvement.

flects genuine cross-LLM variation in structurally complex attack patterns.

Table 12 decomposes DD Full F1 into true positive and false positive rates. TP rates are identical regardless of the benign source: domain-adapted models detect Llama-generated jailbreaks at the same rate whether evaluated against Qwen or Llama benign text. The F1 collapse in the full Llama condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text (FP rate: 88–90% on Llama vs. 0–0.9% on Qwen for adapted variants). The model implicitly treats non-Qwen text style as a jailbreak indicator, a spurious correlation from single-LLM training data. This failure reflects a data diversity limitation, not a methodological one; mixed-LLM training (Table 13) partially addresses this confound.

I Mixed-LLM Training Results

Table 13 reports detection performance when training on a mixed corpus of Qwen3-8B and Llama-3.1-8B conversations. Classification-based DAPT (9-class) maintains the strongest DD OOD performance (0.818±0.096), outperforming both vanilla (0.291±0.065) and MLM (0.289±0.101) by 53 pp. ActorAttack remains challenging under mixed-LLM conditions (9-class F1 = 0.504±0.107), suggesting structurally complex attacks require larger or more diverse training data. ToxicChat performance shows no significant improvement over single-source training (0.550±0.052 vs. 0.596±0.087; Section 4.6), consistent with ToxicChat being zero-shot single-turn transfer rather than a standard OOD benchmark.

J Dataset Scale Validation

Table 14 validates robustness to dataset scale by doubling the training corpus from 500 to 1,000 conversations with new topic seeds. The hierar-

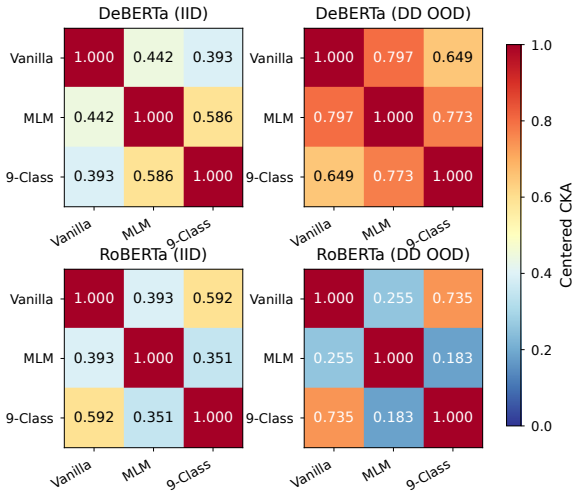


Figure 8: Centered CKA similarity (Kornblith et al., 2019) between encoder variants on IID and OOD data for DeBERTa-v3 and RoBERTa. Low off-diagonal values (vanilla vs. adapted variants) confirm that continued pretraining substantially reshapes representations; the MLM vs. 9-class block shows distinct geometries (CKA ≤ 0.2 on RoBERTa OOD).

Table 11: Cross-LLM validation (F1, full conversation). All models are trained on Qwen3-8B data; conditions vary the source of jailbreak (JB) and benign (ben.) conversations at inference. Values are evaluated under a matched 50-sample subset protocol and differ from Table 3, which uses the full evaluation set.

	DD Full			*AA Full*		
Condition	Van.	MLM	9-cl.	Van.	MLM	9-cl.
Qwen JB + Qwen ben.	.312	.914	.997	.258	.893	.972
Llama JB + Llama ben.	.261	.282	.433	.261	.368	.321
Llama JB + Qwen ben.	.302	.778*	1.000	.302	.903	.801

*High seed variance (std = 0.306): seed 456 outlier (F1 = 0.345); remaining two seeds achieve F1 $\geq$ 0.97.

chy 9-class $\gg$ MLM $\approx$ vanilla holds at 2$\times$ scale: classification-based DAPT maintains 40–56 pp advantage over vanilla, while MLM provides no improvement over vanilla on either attack family (DD: 0.338 vs. 0.326; AA: 0.351 vs. 0.508).

The performance decline at 1,000 conversations may reflect distributional heterogeneity introduced by diverse topic seeds: the additional conversations broaden the training distribution, potentially diluting the concentrated attack-relevant features that smaller, more homogeneous datasets preserve. IID evaluation was precluded by train/test overlap in the expanded dataset (71%), limiting analysis to OOD metrics. Human-data validation also degrades under scaled training (ToxicChat F1: 0.596 $\rightarrow$ 0.341), suggesting the expanded distribution shifts representations away from features useful for both synthetic OOD and human data.

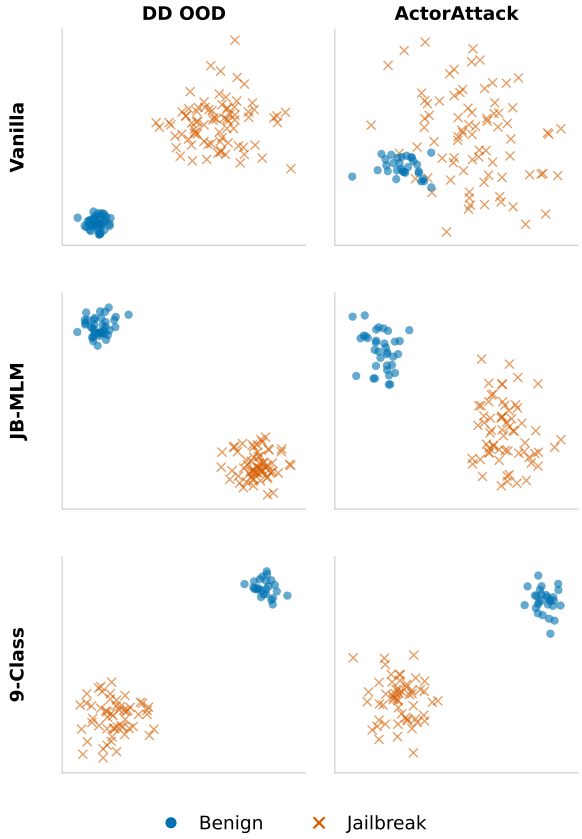


Figure 9: t-SNE visualization of mean-pooled [CLS] embeddings (perplexity=30) for Vanilla, JB-MLM, and 9-Class variants on DD OOD and ActorAttack (118 samples each; blue=benign, red=jailbreak).

K RoBERTa-large DAPT Learning Rate Control

Reducing the DAPT learning rate from 2×10^{-5} to 1×10^{-5} worsens both DD and AA performance, which rules out over-adaptation as the cause of the regression observed in Table 4. An MLM-only variant (without classification) produces the same pattern (AA Full = 0.874$\pm$0.014); representation drift from continued pretraining, not classification-specific bias, is the underlying cause. Domain-adaptive fine-tuning narrows the representation space of a sufficiently large encoder and reduces its inherent OOD generalization ability.

L TOST Equivalence Sensitivity Analysis

In our 80-sample test set, $\delta = 5$ pp corresponds to $\approx$4 additional missed attacks. We choose $\delta = 5$ pp as a conservative threshold; equivalence holds even at $\delta = 3$ pp on DD ($p < 0.001$, $n = 10$) and FITD ($p \approx 0$). With 10 seeds at $\delta = 5$ pp, DD

Table 12: Error decomposition for DD Full cross-LLM conditions (matched 50-sample subsets). TP rate is identical across benign conditions; the F1 collapse in the Llama-only condition (Table 11, row 2) is driven entirely by false positives on unfamiliar benign text.

Metric	Van.	MLM	9-cl.
TP rate (Llama JB)	0.0%	68.0%	100.0%
FP rate (Llama ben.)	29.3%	88.0%	90.0%
FP rate (Qwen ben.)	0.0%	0.9%	0.0%

Table 13: Mixed-LLM training (250 Qwen + 250 Llama conversations, 3-seed mean±std where available). ToxicChat evaluated on balanced set (408 conversations).

Variant	Mixed	DD OOD	AA OOD	FITD OOD	ToxicChat
9-class	**.996±.006**	**.818±.096**	**.504±.107**	**.997±.005**	**.550±.052**
MLM	.681±.226	.289±.101	.258±.075	.679±.188	.256±.053
Vanilla	.802±.010	.291±.065	.389±.073	.867±.022	.356±.009

Table 14: Dataset scale validation (1,000 Qwen conversations, 3-seed mean±std). IID results omitted due to train/test overlap (71% of original test set present in expanded training data).

Variant	DD OOD	AA OOD
9-class	**.889±.041**	**.913±.041**
MLM	.338±.010	.351±.018
Vanilla	.326±.000	.508±.085

Table 15: RoBERTa-large (355M) DAPT learning rate control on OOD Full (3-seed mean±std). Reducing the learning rate worsens both DD and AA, ruling out over-adaptation as the cause of the regression in Table 4.

Config	DD Full	AA Full
Vanilla (no DAPT)	.927±.036	1.000±.000
9-class, LR=2×10^{-5}	.987±.004	.946±.057
9-class, LR=1×10^{-5}	.948±.034	.781±.054

and FITD confirm formal equivalence ($p < 10^{-8}$, $|d| \leq 0.14$), demonstrating that label correctness is not a detection factor for template attacks. On ActorAttack, the mean gap is 3.0 pp with moderate effect size ($d = 0.44$); TOST does not confirm equivalence ($p = 0.183$), indicating that strategy-diverse attacks retain some sensitivity to label quality.

M Partition Robustness and Topic Failure Analysis

M.1 Random Binary Partition Robustness

All 10 partitions achieve perfect IID and DD OOD detection regardless of how the 9 strategy labels are split into two groups. On ActorAttack, 9/10 partitions achieve F1 = 1.000; the sole outlier (partition 7, AA F1 = 0.636) uses a 4/5 split that isolates a specific strategy cluster. The near-universal success confirms that domain-relevant binary classification—not specific label semantics—drives cross-attack generalization on template attacks. The AA outlier is consistent with the complexity-dependent boundary observed in Section 4.3: structurally complex attacks are sensitive to specific partition choices while template attacks are not.

M.2 Topic Classification Failure Analysis

Topic DAPT achieves *higher* per-turn Fisher separability than 9-class DAPT on IID and DD (0.653 vs. 0.200 on DD) yet produces far worse conversation-level detection (F1 = 0.541 vs. 0.994). This rules out per-turn separability destruction as the failure mechanism. The 9-class model instead greatly expands the embedding space (centroid L_2 distance 7.5–27.6 vs. 0.9–1.1 for vanilla) while distributing both classes broadly, creating richer representation geometry that the downstream GRU exploits over the turn sequence. Topic classification teaches features that separate turns individually but are temporally inconsistent across a conversation—high per-turn separability does not guarantee that the GRU can learn a stable conversation-level decision boundary.

N Reproducibility Details

Stage-1 training. All variants use DeBERTa-v3-base (184M parameters) with learning rate 2×10^{-5}, AdamW optimizer (β_1=0.9, β_2=0.999, weight decay 0.01), linear warmup over the first 10% of steps, batch size 16, and 4 epochs. MLM variants use 15% random token masking. Training takes approximately 30 minutes per variant on a single NVIDIA A100 (40GB).

Stage-2 training. The BiGRU has hidden dimension 256, 2 layers, and dropout 0.3 (bidirectional, so output dimension 512). The MLP classifier has layers $512 \rightarrow 256 \rightarrow 2$ with ReLU activation. Training uses learning rate 1×10^{-3}, Adam optimizer, batch size 32, and 20 epochs with early stopping (patience 5) on validation F1. Stage-2 training completes in under 5 minutes per variant.

Data generation. Jailbreak conversations are generated using Qwen3-8B with attack-specific prompt templates. Crescendo conversations follow

Table 16: TOST equivalence test (scrambled vs. 9-class, δ=5 pp, 10 seeds). $p < 0.05$ confirms equivalence within $\pm\delta$.

Dataset	p	d	Equivalent?
DD OOD	2.06×10^{-9}	-0.136	✓
FITD OOD	≈ 0	0.000	✓
ActorAttack	.183	0.440	–

Table 17: OOD F1 for 10 random binary partitions of the 9 strategy labels (each partition randomly assigns strategies to group A or B). All partitions use identical Stage-1 DAPT and Stage-2 GRU training; only the binary label mapping differs. Seed 42 (3 runs).

Partition	Group A	IID	DD OOD	AA OOD
0	{0,2,4,5,7}	1.000	1.000	1.000
1	{1,3,5,6,8}	1.000	1.000	1.000
2	{0,1,3,7,8}	1.000	1.000	1.000
3	{2,4,5,6,7}	1.000	1.000	1.000
4	{0,1,2,6,8}	1.000	1.000	1.000
5	{0,3,4,7,8}	1.000	1.000	1.000
6	{1,2,3,5,6}	1.000	1.000	1.000
7	{1,4,6,7}	1.000	1.000	0.636
8	{0,2,3,4,8}	1.000	1.000	1.000
9	{0,1,5,7,8}	1.000	1.000	1.000
Mean±Std		1.000±.000	1.000±.000	0.964±.109

a 5–8 turn gradual escalation structure. FITD conversations use a 2–3 turn commitment-consistency structure. Deceptive Delight conversations embed harmful content within emotionally engaging 3–5 turn sequences. Benign conversations are generated with topic-anchored prompts across five domains (technology, health, finance, education, general) with matched turn counts.

Strategy annotation. Each user turn is annotated by Qwen3-8B using a structured prompt that provides definitions and two examples for each of the nine strategy categories. The prompt instructs the model to output a single category label. All generation and annotation prompts, the full dataset, and training code are available at https://anonymous.4open.science/r/persuasion_turn_jailbreak_detect-C207/.

Adversarial paraphrasing. The paraphrased test set is generated by instructing Qwen3-8B to rephrase each user turn while preserving the communicative intent. The resulting paraphrases have mean BERTScore 0.92 and mean lexical overlap 0.20 with the originals. These metrics confirm substantial surface-level variation.

Cross-checkpoint variance. To disentangle Stage-1 from Stage-2 variance, we evaluate three independently trained 9-class DAPT checkpoints (seeds 42, 123, 456) with a fixed downstream BiGRU seed (42). Results: IID F1 = 1.000±0.000, DD OOD = 0.926±0.026, AA OOD = 0.935±0.013. Stage-1 training is stable (OOD std $\leq$ 2.6 pp), confirming that the variance reported in the main paper is not dominated by Stage-1 initialization.

Table 18: Per-turn Fisher ratio (mean across all dimensions) and silhouette score between jailbreak and benign turn embeddings ([CLS] representations). Higher values indicate greater per-turn separability.

Model	Dataset	Fisher (mean)	Silhouette
Vanilla	IID test	0.118	0.034
Vanilla	DD OOD	0.401	0.411
Vanilla	AA OOD	0.408	0.330
9-class DAPT	IID test	0.073	0.048
9-class DAPT	DD OOD	0.200	0.079
9-class DAPT	AA OOD	0.832	0.301
Topic DAPT	IID test	0.108	0.076
Topic DAPT	DD OOD	0.653	0.274
Topic DAPT	AA OOD	0.311	0.141

O End-to-End Fine-Tuning Upper Bound

Table 19: Frozen probing vs. end-to-end (E2E) fine-tuning for DeBERTa-v3-base 9-class (3-seed mean±std). E2E unfreezes the Stage-1 encoder during Stage-2 training. Frozen values use main-experiment checkpoints (matching Tables 2–3). Table 25 independently verifies these values.

Config	IID	DD OOD	AA OOD
Frozen probing	1.000±.000	.997±.005	.972±.026
E2E fine-tuning	1.000±.000	1.000±.000	**.997±.005**
Δ	.000	+.003	+.025

End-to-end fine-tuning unfreezes the Stage-1 encoder during Stage-2 BiGRU training, allowing gradient flow through the full pipeline. On IID and OOD DD, the two configurations are indistinguishable; on structurally complex ActorAttack, E2E recovers +2.5 pp. The modest gain confirms that frozen probing captures the vast majority of discriminative information encoded by Stage-1 DAPT, and that the remaining OOD gap in the frozen setting reflects downstream optimization constraints rather than missing representational capacity.

P Real-World False Positive Rate

Table 20 evaluates false positive rates on 226 WildChat benign multi-turn conversations from real

Table 20: False positive rate (%) on 226 WildChat benign multi-turn conversations (GPT-3.5 users, avg 7.4 turns per conversation). All models trained on synthetic Qwen3-8B data (1,000-conversation scale). 3-seed mean±std.

Variant	Seed 42	Seed 123	Seed 456	Mean±Std
9-class	66.8	61.1	76.5	68.1±6.4
MLM	36.7	64.6	32.3	44.5±14.3
Vanilla	47.3	42.5	54.4	48.1±4.9

Table 21: OASST2 benign adaptation curve: FPR on 200 OASST2 benign conversations and OOD jailbreak detection F1 as a function of N OASST2 conversations added to training. Seed 42. Compare WildChat (Table 20): FPR reaches 0% at $N = 10$.

N	FPR (%)	IID F1	DD F1	AA F1	FITD F1
0	57.5	–	–	–	–
10	20.5	1.000	1.000	1.000	1.000
25	**1.5**	1.000	1.000	0.953	1.000
50	**0.5**	1.000	1.000	0.990	1.000

GPT-3.5 users (avg 7.4 turns). All variants produce FPR far exceeding deployable thresholds: 9-class is worst at 68.1%, followed by vanilla (48.1%) and MLM (44.5%). Domain adaptation *increases* FPR relative to vanilla (+20 pp for 9-class), indicating that continued pretraining amplifies dependence on the training benign distribution. This is consistent with the cross-LLM false positive spike (Table 12): models trained on synthetic Qwen3-8B benign text implicitly treat unfamiliar conversational styles as jailbreak indicators. High MLM seed variance (std = 14.3%) suggests that unsupervised pretraining is particularly unstable under benign distribution shift. Deployment requires per-domain benign calibration or mixed-source benign training data.

Post-hoc calibration. ROC AUC = 0.700±0.054 indicates insufficient discrimination between WildChat benign and jailbreak conversations at the representation level. At TPR≥95%, the optimal threshold still yields FPR = 62.7%, far above deployable levels. Temperature scaling cannot improve this trade-off, as ROC curves are invariant to monotonic score transforms. The AUC ceiling reflects genuine domain shift: WildChat benign and jailbreak representations overlap substantially, and no post-hoc calibration can overcome this; in-domain benign adaptation is required.

Second benign source: OASST2. To test whether benign-side shift generalizes beyond WildChat, we evaluate on 200 benign multi-turn conversations from OpenAssistant OASST2 (Köpf et al., 2023)—a crowdsourced dataset of real human conversations, distinct from WildChat (GPT-3.5 user logs) in both authorship and topic distribution.

Baseline FPR on OASST2 (57.5%) is comparable to WildChat (68.1%), confirming that benign-side shift is not source-specific but a general consequence of synthetic-only benign training. The adaptation curve follows the same pattern—monotonic FPR reduction with preserved OOD detection—but requires more samples: 25 conversations reduce FPR to 1.5% (vs. 0% at $N = 10$ for WildChat), and 50 conversations reach 0.5%. The slower convergence likely reflects greater distributional distance between crowdsourced OASST2 conversations and the synthetic training distribution compared to WildChat's GPT-3.5 user logs.

P.1 Estimated Dual-OOD Detection Performance

The OOD detection F1 reported in the main paper evaluates attack-side OOD with IID benign examples (FPR = 0), providing an upper bound. To estimate performance when *both* attack and benign inputs are OOD, we combine per-seed attack recall (from OOD evaluation) with per-seed benign FPR (from WildChat/OASST2 experiments), assuming independence between attack detection and benign false positives. For each seed, we compute binary detection F1 from the resulting TP, FP, FN counts on the OOD test composition (80 jailbreak + 38 benign).

Table 22: Estimated dual-OOD binary detection F1 under different benign conditions (3-seed mean±std). Attack recall held fixed from OOD evaluation; only benign FPR varies. IID row corresponds to the benign-IID upper bound (FPR = 0).

Benign condition	DD OOD F1	AA OOD F1
IID (FPR = 0)	.994±.009	.981±.019
WildChat, no adapt	.855±.020	.842±.012
OASST2, no adapt	.874±.009	.861±.018
WildChat + 10-shot	.994±.009	.981±.019
OASST2 + 25-shot	.991±.009	.978±.018

Without benign adaptation, dual-OOD F1 drops by 13–14 pp (DD: .994 → .855; AA: .981 → .842), driven entirely by false positives on OOD benign inputs. After few-shot adaptation—10 WildChat or 25 OASST2 conversations—dual-OOD F1

recovers to within 0.3 pp of the IID upper bound, confirming that the lightweight adaptation procedure (Section 4.8) effectively closes the benign distribution gap without compromising attack-side performance.

Q BERT-base Proxy Baseline

Table 23: BERT-base-uncased (110M) frozen probing without DAPT, matching DeepContext's encoder family (Albrethsen et al., 2026). 3-seed mean±std. DeBERTa-v3 results from Table 4 for comparison.

Encoder	IID	DD OOD	AA OOD
BERT-base (110M), no DAPT	.996±.006	.318±.031	.275±.027
DeBERTa-v3 (184M), no DAPT	1.000±.000	.312±.043	.258±.000
DeBERTa-v3 (184M), 9-class	1.000±.000	.994±.005	.939±.047

Table 23 compares BERT-base-uncased (110M parameters, matching DeepContext's encoder) against DeBERTa-v3-base (184M), both without domain-adaptive pretraining. The two vanilla encoders achieve comparable OOD performance (DD: 0.318 vs. 0.312; AA: 0.275 vs. 0.258), despite a 1.7× parameter difference and different pretraining procedures. Both collapse on OOD evaluation while 9-class DAPT recovers to 0.994 (DD) and 0.939 (AA). This confirms that cross-attack generalization is driven by domain adaptation, not encoder architecture or capacity at this scale.

R Confidence Calibration

Table 24: Expected calibration error (ECE, 10 bins) for 9-class and vanilla on IID and OOD evaluation (3-seed mean±std). Lower is better.

Variant	IID	DD OOD	AA OOD
9-class	≈0.000	0.143±0.053	0.116±0.053
Vanilla	0.009±0.007	0.669±0.004	0.504±0.066

The 9-class model achieves substantially lower ECE than vanilla on OOD data (0.12–0.14 vs. 0.50–0.67), indicating that domain adaptation improves not only discriminative performance but also confidence calibration. IID calibration is near-perfect for both variants. The residual OOD ECE (0.12–0.14) suggests that temperature scaling could further improve deployment reliability.

R.1 ToxicChat False Positive Confidence

On ToxicChat, 9-class false positives are overwhelmingly high-confidence: 70.2% of majority-vote FPs exceed $p > 0.9$ (median 0.999), compared to 58.8% for JB-MLM (median 0.933). Per-seed analysis is more extreme—94–98% of single-model 9-class FPs are high-confidence (median 1.0), with the borderline fraction (25%) in majority vote arising from cross-seed disagreement rather than individual model uncertainty. This confirms that the 9-class ToxicChat FPR reflects systematic feature misalignment—classification DAPT shifts benign text into the jailbreak region with high confidence—rather than boundary ambiguity.

S Frozen Probing Verification

Table 25: Frozen probing vs. end-to-end (E2E) fine-tuning (3-seed mean±std). 9-class: primary DAPT checkpoints; MLM: matched MLM control checkpoints. Frozen probing trains only BiGRU+MLP on frozen DAPT embeddings.

Config	IID	DD OOD	AA OOD	FITD OOD
Frozen 9-class	1.000±.000	.997±.005	.972±.026	1.000±.000
Frozen MLM	.982±.016	.993±.009	.941±.037	.987±.009
E2E 9-class	1.000±.000	1.000±.000	.997±.005	—

Table 25 compares frozen probing across DAPT objectives and against end-to-end fine-tuning. On easy OOD (DD, FITD), frozen MLM and frozen 9-class representations perform comparably ($\Delta \leq 1.3$ pp). On hard OOD (ActorAttack), 9-class retains a modest advantage (0.972 vs. 0.941, $\Delta = 3.1$ pp) with lower variance ($\sigma = 0.026$ vs. 0.037). Both far exceed vanilla encoders ($\leq$0.318; Table 23). The E2E advantage over frozen 9-class is $\leq$2.5 pp, confirming that Stage-1 representation quality is the primary source of OOD generalization.

A striking asymmetry emerges when comparing frozen probing to the full pipeline (Table 3): MLM frozen representations *outperform* the full MLM pipeline on AA OOD (0.941 vs. 0.893), while 9-class frozen exactly matches its full pipeline (0.972 vs. 0.972). This indicates that MLM's cross-seed instability originates in Stage-2 BiGRU training rather than Stage-1 DAPT: the MLM encoder produces adequate representations that are then degraded by unstable downstream optimization. Classification DAPT, by contrast, produces representations that remain robust through Stage-2 training, explaining its consistently lower variance (Table 3).

This two-level structure (DAPT vs. no-DAPT producing a representation quality leap of +70 pp; classification vs. MLM DAPT producing a stability difference) is consistent with two further observations: λ=0 matches $\lambda > 0$ during E2E fine-tuning

(Table 35), and mean pooling matches BiGRU under E2E (Table 30). Once DAPT representations are strong, downstream configuration details have minimal impact.

S.1 Single-Turn Truncation Analysis

To isolate whether the ToxicChat evidence-hierarchy reversal (Section 5) reflects multi-turn structural dependence, we truncate each synthetic test conversation to its last user turn and re-evaluate both 9-class and JB-MLM models (3-seed mean).

Table 26: Full-conversation vs. last-turn-only F1 (3-seed mean). Truncation widens the 9-class advantage, ruling out multi-turn structure as the driver of the ToxicChat reversal.

Dataset	Mode	9-class	JB-MLM
IID	Full	1.000	.983
IID	Last turn	.991	.894
DD OOD	Full	.997	.993
DD OOD	Last turn	.942	.924
AA OOD	Full	.972	.941
AA OOD	Last turn	.978	.927
FITD OOD	Full	1.000	.987
FITD OOD	Last turn	1.000	.927

Mean Δ (9-class $-$ MLM): Full +1.6 pp, Last turn +6.0 pp. 9-class advantage widens under truncation across all families, indicating the reversal on ToxicChat is driven by distributional differences between synthetic and human text, not by multi-turn structural dependence.

Representation geometry analysis. To characterize the mechanism behind classification DAPT's robustness, we analyze Stage-1 representations across all seven DeBERTa-v3-base variants (seed 42). We extract [CLS] embeddings from 150 test conversations (50 IID + 50 DD OOD + 50 AA OOD) and compute three metrics.

Classification variants compress representations into 2–9 effective dimensions (vs. 25–32 for MLM/vanilla) while increasing Fisher ratio by 3–5 orders of magnitude. Weight distances from pretrained are comparable across all variants ($\sim$0.2–0.4%), indicating that the mechanism is representational restructuring rather than parameter-space regularization. Scrambled labels produce similar compression (PR = 7.8, Fisher = 2.6), confirming that the classification objective structure—not label semantics—drives discriminative compression.

Table 27: Representation geometry and class separability across Stage-1 variants. PR: participation ratio (effective dimensionality of the embedding space). Fisher ratio: inter-class distance^2 / sum of intra-class variances. Weight Δ: L2 distance from pretrained weights (relative to pretrained norm).

Variant	PR	Fisher	Wt. Δ (%)
Vanilla	31.9	$<$0.001	0.000
Wiki-MLM	25.0	$<$0.001	0.221
JB-MLM	24.6	$<$0.001	0.256
Topic	14.5	0.012	0.371
Scrambled	7.8	2.60	0.283
9-class	9.4	4.47	0.303
Binary	3.7	19.7	0.239

Layer-wise localization. To determine *where* discriminative compression occurs, we compute PR and Fisher ratio at every DeBERTa-v3-base layer (0–12) for four variants (9-class, scrambled, JB-MLM, vanilla) using the same 236-sample evaluation set (seed 42).

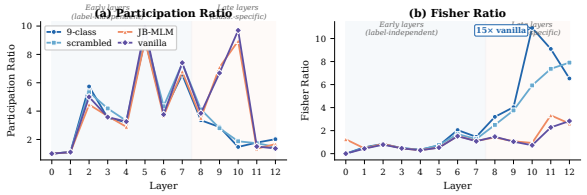


Figure 10: Layer-wise representation analysis. (a) Participation ratio: classification variants (9-class, scrambled) show sharp dimensional collapse at layers 9–10 (PR $\approx$ 1.5–2.9) while vanilla/JB-MLM maintain high dimensionality (PR $\approx$ 7–10). (b) Fisher ratio (log scale): class separability diverges from layer 8, with 9-class peaking at 10.9 (layer 10) vs. vanilla 0.73 (15$\times$).

Layers 0–7 show no systematic divergence across variants (Fisher $<$ 2.1 for all; PR tracks similar oscillation patterns), consistent with Maennel et al. (2020)'s finding that early layers learn label-independent features. From layer 8 onward, classification-supervised variants diverge sharply: 9-class Fisher rises from 3.2 (layer 8) to 10.9 (layer 10, peak), while vanilla remains below 1.5 throughout. Simultaneously, 9-class PR drops from 3.4 to 1.5 at layer 10—a 6.6$\times$ compression relative to vanilla (PR = 9.7). JB-MLM tracks vanilla across all layers (Fisher $\leq$ 3.3, PR $\geq$ 1.4), confirming that MLM does not induce discriminative restructuring. Scrambled labels exhibit the same late-layer pattern (Fisher peaks at 7.9, layer 12; PR drops to 1.5, layer 12), reinforcing that any classification signal—regardless of label semantics—triggers late-layer

compression.

Discriminative subspace probe. To test whether the compressed subspace is *sufficient* for OOD detection, we project layer-10 [CLS] representations onto the top-k SVD components and train a logistic regression probe (3 seeds, sklearn default). Controls: random k-dimensional projections (10 random bases, mean reported) and vanilla/MLM models under identical projection.

Table 28: Discriminative subspace probe: OOD F1 when projecting layer-10 representations onto top-k SVD components vs. random k-dimensional projections. 9-class top-3 achieves perfect AA detection with 3/768 dimensions; MLM fails even with all 768.

Model	**Projection**	DD	AA	FITD
9-class	top-3	1.000	**1.000**	1.000
9-class	top-5	1.000	1.000	1.000
9-class	random-3	1.000	0.928±.129	1.000
9-class	random-5	1.000	0.952	1.000
9-class	full-768	1.000	1.000	1.000
scrambled	top-3	1.000	0.962	1.000
scrambled	random-3	0.981	0.766	0.987
JB-MLM	top-3	1.000	0.618	1.000
JB-MLM	full-768	1.000	0.636	1.000
JB-MLM	random-5	0.685	0.584	0.744

Three findings establish a link between compression and generalization (Table 28). First, 9-class top-3 achieves perfect ActorAttack detection (F1 = 1.000) using only 3 of 768 dimensions—all discriminative information resides in the compressed subspace. Second, random 3-dimensional projections degrade to 0.928 on AA, confirming that the specific SVD directions matter. Third, JB-MLM reaches only 0.636 on AA even with all 768 dimensions, demonstrating that without classification-induced compression, the representation lacks the structure needed for cross-attack generalization. DD and FITD are near-ceiling for all models, making ActorAttack the critical test. SVD directions computed on IID training data only (without OOD test exposure) yield equal or higher F1 than all-data SVD (AA: 0.947 vs. 0.904), confirming the subspace is not an artifact of test-set leakage.

SVD necessity ablation. The probe above establishes *sufficiency*: 3 SVD dimensions carry enough information. To test *necessity*, we project *out* the top-3 SVD directions from full 768-dimensional 9-class [CLS] embeddings and re-run BiGRU classification on OOD data (3 seeds). A random-3 removal control verifies that the effect is specific to the discriminative subspace. The top-3 directions explain 89.8% of training-set variance.

Table 29: SVD necessity ablation: OOD F1 after projecting out top-3 vs. random-3 SVD directions from 9-class layer-10 embeddings (3-seed mean±std).

Condition	DD OOD	FITD OOD	AA OOD
Original 768-dim	1.000±.000	1.000±.000	.994±.005
Remove top-3 SVD	.831±.064	1.000±.000	.938±.011
Remove random-3	1.000±.000	1.000±.000	.994±.005

Removing top-3 SVD directions degrades DD detection by 17 pp (F1: $1.0 \rightarrow 0.831$) and AA by 5.6 pp ($0.994 \rightarrow 0.938$), while random-3 removal has zero effect. FITD is unaffected by either ablation, indicating its discriminative information is more distributed. Combined with the sufficiency result (Table 28), the top-3 SVD directions are both sufficient and necessary for DD detection, sufficient but partially redundant for AA, and sufficient but not necessary for FITD.

T Temporal Aggregation Comparison

Table 30: Temporal aggregation comparison: mean pooling vs. BiGRU under frozen and end-to-end (E2E) settings (3-seed mean±std). Primary 9-class DAPT checkpoints.

Setting	**Aggregator**	DD OOD	AA OOD
Frozen	BiGRU	.997±.005	.972±.026
	Mean pooling	.991±.013	.962±.015
E2E	BiGRU	1.000±.000	.997±.005
	Mean pooling	1.000±.000	1.000±.000

Table 30 compares BiGRU and mean pooling under both frozen and E2E settings. Under frozen probing (the main paper's pipeline), BiGRU provides a marginal advantage on ActorAttack (+1.0 pp) with overlapping confidence intervals, consistent with the main text finding that temporal order is not a core detection factor ($|\Delta| < 1$ pp under turn shuffling; Section 4.3). Under E2E fine-tuning, the difference vanishes entirely: mean pooling saturates all benchmarks at F1 = 1.000. The choice of frozen BiGRU in the main pipeline reflects a design trade-off: it enables representation-level analysis (CKA, t-SNE) that E2E would confound, while sacrificing $\leq$1 pp on the hardest OOD set.

Table 31: Jailbreak recall by conversation length bin on OOD evaluation (3-seed majority vote). Benign conversations span 12–20 turns; jailbreak conversations span 3–10 turns.

Dataset	Variant	1–3 turns	4–6 turns	7–10 turns
DD OOD	9-class	—	100.0%	99.4%
	Vanilla	—	≤2.6%	≤1.7%
AA OOD	9-class	93.3%	95.6%	100.0%
	Vanilla	0.0%	≤2.6%	0.0%

U Length Stratification Analysis

Table 31 stratifies jailbreak recall by conversation length. The 9-class model shows no systematic degradation: ActorAttack recall increases monotonically from 93.3% (1–3 turns) to 100% (7–10 turns), and DD recall is near-perfect across all bins.

Length confound. Benign conversations (12–20 turns) and jailbreak conversations (3–10 turns) have zero length overlap in the OOD evaluation sets, raising the concern that the model discriminates by conversation length rather than content. Three observations argue against this interpretation: (1) Vanilla DeBERTa has access to identical length information via the BiGRU but achieves recall ≤2.6% across all bins; length alone is insufficient without domain-adapted representations. (2) At prefix K=1, where all conversations are length-equalized, 9-class still achieves F1 = 0.773±0.041 on ActorAttack (Table 8), demonstrating content-based detection independent of length. (3) The model is trained on IID data (Crescendo) and cannot learn OOD-specific length distributions; generalization to unseen length patterns must rely on content features that transfer across attack families.

Direct empirical verification. Truncating benign conversations to match jailbreak turn counts (Part A) produces zero false positives across all truncation levels and all 3 seeds, directly confirming that the model does not use conversation length as a detection shortcut. K=3 and K=5 are the effective control conditions where all 38 conversations were truncated; K=10 exceeds the maximum test benign length and serves as a no-op control.

WildChat FPR by turn count (Part B) provides complementary evidence: short conversations (3–10 turns, n=196) have *lower* FPR than medium-length ones (11–15 turns, n=23), inconsistent with a "short = jailbreak" shortcut. Medium and long buckets have limited sample sizes and should be interpreted as directional evidence only.

Table 32: Turn-count independence verification (9-class DeBERTa-v3-base, same checkpoint as Table 2, 3 seeds). Evaluated on the full OOD test set (80 jailbreak + 75 benign conversations) rather than the balanced subset used in Table 2. Part A: FPR on test benign truncated to K user turns (all seeds FPR = 0). Part B: WildChat FPR by turn-count bucket (mean±std). Part C: jailbreak detection (F1) after padding with benign turns to 15+ total turns (mean±std where available).

Condition	FPR / F1	n
Part A: Benign truncation (test set)		
Original (12–20 turns)	FPR = .000	38
Truncated K=3	FPR = .000	38
Truncated K=5	FPR = .000	38
Truncated K=8	FPR = .000	38
Truncated K=10*	FPR = .000	38
Part B: WildChat FPR by turn count		
Short (3–10 turns)	FPR = .770±.105	196
Medium (11–15 turns)	FPR = .942±.054	23
Long (16+ turns)	FPR = .857±.117	7
Part C: Jailbreak padding to 15+ turns		
DD original	F1 = .996	80
DD padded	F1 = .971 (Δ=−2.5 pp)	80
AA original	F1 = .963	80
AA padded	F1 = .383±.261 (Δ=−57.9 pp)	80

*K=10 did not truncate any conversation (test benign max ∼8 user turns).

Padding jailbreak conversations with benign turns (Part C) reveals an asymmetry: DD detection is resilient (Δ=−2.5 pp), but ActorAttack collapses (Δ=−57.9 pp). Padding experiments append benign turns *after* the complete attack sequence; in practice, adversaries would not dilute their own successful attacks. The collapse nonetheless exposes BiGRU temporal aggregation as vulnerable to attention dilution on subtle attack signals, a limitation discussed in Section 6.

V Inference Efficiency

Table 33: Inference efficiency comparison. DeBERTa+BiGRU and TF-IDF+LR measured on IID test (76 conversations, avg 8.3 turns, single A100). LLM latency from separate OOD evaluation (118 conversations); cross-condition comparison is indicative only.

Method	Latency (ms)	Params	VRAM
TF-IDF+LR	0.43±0.05	∼10K	CPU
DeBERTa+BiGRU (ours)	12.1±1.2	186.7M	862 MB
Qwen3-8B 3-shot†	∼5,200	8B	∼16 GB

†Qwen3-8B latency from OOD evaluation (DD/AA, 118 conversations); not directly comparable to encoder latency measured on IID test.

Table 33 compares inference cost. On matched data (IID test), DeBERTa+BiGRU processes a con-

versation in 12.1 ms, 28× slower than TF-IDF (0.43 ms), but this speed gap purchases +71 pp on ActorAttack (F1 = 0.972 vs. 0.261; Table 3). LLM-as-judge approaches are orders of magnitude slower (∼5.2 s per conversation for Qwen3-8B 3-shot) while achieving substantially lower OOD detection (DD = 0.566, AA = 0.652; Table 6). Zero-shot Qwen3-8B achieves F1 = 0.0 on DD OOD because it predicts all conversations as benign (Table 6). The full DeBERTa+BiGRU pipeline requires 862 MB GPU memory, enabling deployment on consumer hardware. The 24.3× speedup reported in the main text derives from a matched-condition comparison (DisPeD 13.12 ms vs. Qwen3-8B 318.53 ms on the same GPU and data); the larger gap in Table 33 reflects different hardware conditions and evaluation sets.

W Data Efficiency

Table 34: Data efficiency of 9-class DAPT (3-seed mean±std, sample std with $n-1$ divisor) and vanilla baseline (seed 42 only). N = total training conversations (balanced jailbreak/benign).

N	**Variant**	IID	DD OOD	AA OOD
50	9-class	1.000±.000	1.000±.000	.969±.037
125	9-class	1.000±.000	1.000±.000	.975±.019
250	9-class	1.000±.000	1.000±.000	.975±.035
350	9-class	1.000±.000	.978±.030	.972±.033
50	Vanilla	.646	.529	.234
125	Vanilla	.532	.244	.244
250	Vanilla	.987	.365	.502
350	Vanilla	1.000	.360	.489

Table 34 shows that 9-class DAPT reaches near-saturation performance with as few as 50 conversations (25 jailbreak + 25 benign): AA OOD F1 = 0.969±0.037, already surpassing vanilla at full training size (N=350, AA = 0.489) by 48 pp using only 14% of the data. DD OOD is perfectly solved at all training sizes. Vanilla performance is noisy across N (single seed); the consistent pattern is that vanilla never exceeds DD = 0.53 or AA = 0.50 regardless of training size, while 9-class maintains AA ≥ 0.969 at all N. Seed 456 is systematically lower across all N values, contributing most of the cross-seed variance; this is consistent with the initialization sensitivity observed in the main results.

Table 35: Effect of auxiliary persuasion loss weight λ during Stage-2 E2E fine-tuning (seed 42). Stage-1 DAPT encoder is identical across all rows; only the Stage-2 loss $L = L_{\text{binary}} + \lambda \cdot L_{\text{strategy}}$ varies.

λ	IID	DD OOD	FITD OOD	AA OOD
0	1.000	1.000	1.000	1.000
0.1	1.000	1.000	1.000	.990
0.5	1.000	1.000	1.000	.981
1.0	1.000	1.000	1.000	.981
2.0	1.000	1.000	1.000	.944

X Auxiliary Loss Weight Ablation

During Stage-2 E2E fine-tuning, the auxiliary persuasion loss weight λ has limited impact on detection (Table 35): the maximum AA OOD gap across all λ values is 5.6 pp (single seed), comparable to the cross-seed variance (∼6 pp) observed in the main results. All configurations converge to perfect validation F1 within epoch 1, indicating that DAPT representations already encode sufficient discriminative features for downstream classification (Section S). The value of persuasion strategy classification lies in Stage-1 DAPT, where it shapes encoder representations that transfer across attack families, not in the Stage-2 auxiliary loss.

Y OOD Strategy Prediction Analysis

Table 36: OOD strategy prediction accuracy (jailbreak turns only) and detection rate for the 9-class model (3-seed mean). r: Pearson correlation between per-strategy prediction accuracy and per-strategy detection rate.

Dataset	Strat. Acc.	Det. Rate	Missed (per seed)	r
DD OOD	50.1%	99.6%	0 / 0 / 1	∼0.45
AA OOD	50.1%	96.2%	1 / 7 / 1	∼0.24

Table 36 reports OOD strategy prediction accuracy alongside detection rates. Strategy prediction accuracy is substantially above random (50.1% vs. 11.1% chance for 9 classes) and positively correlated with detection success ($r \approx 0.24$–0.45), but strategy prediction is not the detection mechanism: frozen probing with a randomly initialized strategy head (Appendix S) achieves 97.2% AA OOD detection, and λ=0 (no strategy loss) matches or exceeds $\lambda > 0$ during E2E fine-tuning (Table 35). The classification objective shapes representations during Stage-1 DAPT training; at inference, strategy predictions are a byproduct of those representations, not a driver of detection.

Z Cross-Seed Ensemble

Table 37: Majority vote ensemble (3 seeds) vs. single-seed mean on OOD Full. "Fixed" = samples corrected by ensemble; "Broke" = samples corrupted by ensemble.

Variant	DD single→ens.	AA single→ens.	Fixed	Broke
9-class	.997→**1.000**	.972→**.981**	5	0
MLM	.914→**1.000**	.897→**.953**	26	0
Vanilla	.258→.258	.258→.258	0	0

Table 37 shows that majority vote never corrupts correct predictions (Broke = 0 for all variants). MLM benefits most from ensembling (+8.6 pp DD, +6.0 pp AA), because its per-seed errors are complementary: each seed fails on different samples, and the majority corrects these. This further exposes MLM's seed instability: its individual seeds are unreliable, but their errors are uncorrelated. By contrast, 9-class seeds are already highly consistent (std ≤ 0.026), leaving minimal room for ensemble improvement (+0.3–0.9 pp). Vanilla errors are systematic (all seeds fail on the same OOD samples), so ensembling provides no benefit.

Z.1 Seed-Level F1 Comparisons

To verify that cross-seed pooling does not mask inconsistent pairwise orderings, Table 38 reports per-seed F1 for all variants on OOD DD and ActorAttack at full conversation length. The overall evidence hierarchy is maintained across seeds, with 9-class and binary consistently outperforming MLM; scrambled shows one reversal on AA (seed 42).

Table 38: Per-seed F1 on OOD Full for key variants (seeds 42, 123, 456). The relative ordering is consistent across seeds.

	DD-Full			AA-Full		
Variant	s42	s123	s456	s42	s123	s456
9-class	1.000	1.000	.990	.990	.935	.990
Binary	1.000	1.000	1.000	.981	.990	1.000
Scrambled	.990	1.000	1.000	.891	.990	.971
MLM-only	1.000	.990	.751	.971	.925	.784
Vanilla	.285	.305	.348	.258	.258	.258

Use of AI Assistance

We used Claude for coding, code drafting, debugging, and language polishing.